

Firm Location, Industrial Base, and the Gains from Trade

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Abstract

I develop a framework in which firms choose where to produce, making each country's industrial base – the mass and productivity distribution of producers located there – an equilibrium object. I embed this mechanism in canonical trade models and show that changes in the industrial base are a first-order determinant of the gains from trade. Using data from 1992 to 2019, I find that this channel is as large as the standard domestic-expenditure-share sufficient statistics and reverses the sign of welfare changes for many countries. In a counterfactual undoing China's WTO-accession productivity growth, endogenous relocation overturns welfare effects for many trading partners.

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1 Introduction

The concern that globalization may erode a country’s industrial base has been at the center of policy debates around the world. Over the past few decades, the expansion of international trade has coincided with a large-scale relocation of global production and, more recently, with a resurgence of industrial policy.¹ Yet little is known about how firm location choices shape the relationship between changes in a country’s industrial base and its gains from trade.

This paper develops a quantitative model of the world economy in which firms choose where to produce. The model treats the geography of production as an equilibrium outcome and uses this margin to study how globalization reallocates firms across countries and how changes in the domestic industrial base mediate the overall gains from trade. The central finding is that the welfare consequences of globalization depend not only on how much a country trades, but also on how trade reshapes its production base. A country may import more and consume cheaper foreign goods, yet gain less or lose, if globalization erodes its domestic production base. Conversely, a country may import less yet gain from trade if this is induced by an expanding industrial base rather than reduced access to foreign goods.

To fix ideas, start from the welfare formula in [Arkolakis et al. \(2012\)](#) based on the domestic-expenditure-share sufficient statistics, π_{ii} , written before imposing the restriction that the country-specific supply shifter, or aggregate technology term, T_i , is fixed. Let Δ denote changes between two equilibria such that changes in welfare can be written as:

$$\Delta \ln W_i = -\frac{1}{\eta} \Delta \ln \pi_{ii} + \Delta \ln T_i,$$

where η is the trade elasticity parameter. In the canonical environments, T_i is treated as a primitive and held fixed, so $\Delta \ln T_i = 0$. The ACR term, $-(1/\eta)\Delta \ln \pi_{ii}$, then summarizes the gains from trade. Under endogenous firm location, this restriction no longer applies and the aggregate term T_i reflects the mass and productivity distribution of firms that choose to locate in country i forming its industrial base. Changes in the latter have direct welfare effects: firms located at home do more than supply the domestic market, they raise the country’s effective aggregate productivity and hence

¹For discussions of global production relocation, see [Hanson \(2012\)](#); for the recent resurgence of industrial policy, see [Juhász et al. \(2024\)](#).

real income.

The key feature of the proposed framework is that it characterizes the productivity distribution of firms located in a given country as a tractable conditional distribution induced by firms' location choices. Firms choose locations to maximize profits, which depend on location-specific productivity draws from multivariate extreme-value or power-law distributions, production costs, and market access. I provide conditions under which this problem can be embedded in different market structures, including perfect competition as in [Eaton and Kortum \(2002\)](#), Bertrand competition as in [Bernard et al. \(2003\)](#), and monopolistic competition in the spirit of [Melitz \(2003\)](#). Across these environments, the equilibrium system takes a common form. The framework therefore remains consistent with canonical models of international trade while changing the interpretation of firm heterogeneity within each country: the productivity distribution of active firms is itself an equilibrium object. Hence, an additional contribution of this paper is providing a microfoundation for location-specific firm heterogeneity as the outcome of selection across production locations.

Next, I use the general framework to derive a sufficient-statistics formula for the welfare gains from trade. The formula shows that the ACR term, $-(1/\eta)\Delta \ln \pi_{ii}$, and the industrial-base term, $\Delta \ln T_i$, may reinforce each other, but may also move in opposite directions. The net welfare effect is therefore a quantitative question and depends on the observed sufficient statistics and on two parameters: the trade elasticity, which governs the strength of the ACR channel, and the firm mobility elasticity, which governs how strongly firms reallocate across locations in response to economic shocks. I structurally estimate both parameters and apply the resulting sufficient-statistics formula to global trade data over 1992-2019.

Quantitative results show that the location margin is a first-order component of the gains from trade. Considering changes between 1992 and 2019, the industrial-base term is approximately as large as the ACR term, and the two terms are negatively correlated across countries: countries whose domestic expenditure shares fall are also more likely to lose industrial base due to globalization. Accounting for this margin changes measured welfare gains substantially and reverses the sign of the welfare change for 25%-42% of countries. I also consider a counterfactual exercise in the spirit of [Eaton and Kortum \(2002\)](#) and [Caliendo et al. \(2018\)](#), which quantifies how productivity changes in one location affect other countries through trade. With endogenous

location choice, reversing China’s aggregate technology primitives to its pre-WTO-accession level induces firms to leave China for other production locations. This relocation response overturns the standard conclusion about the benefits of foreign technology, turning welfare losses into gains for many of China’s trading partners.

Relation to the Literature.

This paper is related to the literature on the *delocation* channel, through which changes in trade costs and policy affect welfare by reallocating production across countries. This tradition includes work on the home-market effect, which links production location to market size (Krugman 1980; Helpman and Krugman 1985), and work on how trade policy reallocates production across borders (Venables 1985, 1987). The welfare results in this paper also speak to the new economic geography literature, which studies how trade liberalization reshapes the spatial distribution of industrial production (Krugman and Venables 1995; Baldwin and Robert-Nicoud 2000; Baldwin et al. 2003).²Relative to this literature, I bring the delocation margin into quantitative trade by developing a unified endogenous firm-location problem that can be embedded in canonical gravity models. The framework preserves tractability and avoids common limitations, such as multiple equilibria, that make older economic geography models difficult to use for counterfactual analysis.

In terms of the broad research question, the paper relates to recent quantitative work that studies the interaction between trade policy, industrial including homogeneous-firm environments (Ossa 2011; Lashkaripour and Lugovskyy 2023) and on heterogeneous-firm models in the spirit of Melitz (2003) and Chaney (2008) as in Haaland and Venables (2016); Spearot (2016); Caliendo et al. (2015, 2023); Caliendo and Parro (2019); Bagwell and Lee (2020) and others. The key distinction is that firms do not enter countries through separate free-entry conditions. Instead, each firm is characterized by a vector of productivities, one for each potential production location, and chooses where its cost-adjusted productivity and market access yield the highest profit. The within-country productivity distribution is therefore itself an equilibrium object, determined by how many firms select into each country and which firms select

²The paper also connects to empirical work on how globalization and trade policy shape the location of industrial activity (e.g., Hanson 1996, 2001; Flaaen et al. 2020), and offers a structural interpretation of why improved market access in one country can erode the industrial base elsewhere, the mechanism behind the manufacturing and local-labor-market effects documented in the China-shock literature (e.g., Autor et al. 2013).

there.

On the modeling side, the paper contributes to a small literature on discrete location choice by heterogeneous firms. It generalizes [Suárez Serrato and Zidar \(2016\)](#) and [Fajgelbaum et al. \(2019\)](#), who study firm location in monopolistically competitive environments with independent location-specific productivity draws, in two directions.³ First, I allow for multivariate extreme-value and power-law productivity distributions to be correlated across locations. Second, I generalize the location problem so that it can be embedded in multiple market structures. Both generalizations matter. Under perfect and Bertrand competition, independence makes the location margin disappear in the aggregate: the gain from attracting more firms is exactly offset by lower average productivity among selected firms, and the model collapses to the canonical [Eaton and Kortum \(2002\)](#) and [Bernard et al. \(2003\)](#) allocations. This helps explain why much of the canonical quantitative trade literature has operated without this margin. My empirical results point to strong cross-location dependence in firm productivity magnifying the role of location margin in shaping the gains from trade across all market structures.

This paper provides an alternative to models in which complementarities and agglomeration forces shape the allocation of firms and workers across space. These include assignment models with supermodularity between agent and location characteristics (see [Costinot and Vogel 2010](#); [Behrens et al. 2014](#); [Sampson 2014](#); [Gaubert 2018](#)), models of spatial sorting with agglomeration economies ([Lucas and Rossi–Hansberg 2002](#); [Baldwin and Okubo 2006](#); [Okubo et al. 2010](#); [Combes et al. 2012](#)), and recent work on how agglomeration and scale affect international trade outcomes (see [Kucheryavyy et al. 2023](#); [Bartelme et al. 2025](#)). The mechanism developed here does not rely on supermodularity or agglomeration externalities. Instead, the paper combines payoff-based discrete choice with multivariate productivity distributions, preserving tractability for both location shares and the productivity distributions induced by selection.⁴

³For empirical evidence on the connection between discrete-choice shares and average productivity of selected firms, also see [Malgouyres et al. \(2023\)](#) and [Suárez Serrato and Zidar \(2023\)](#).

⁴This methodological connection relates the paper to tractable discrete-choice models following [McFadden \(1974, 1978\)](#) and to applications that use discrete choice in spatial, trade, and development settings (among many others, [Artuç et al. 2010](#); [Grogger and Hanson 2011](#); [Allen and Arkolakis 2014](#); [Dix-Carneiro 2014](#); [Ahlfeldt et al. 2015](#); [Holmes and Sieg 2015](#); [Monte et al. 2018](#); [Bryan and Morten 2019](#); [Fajgelbaum and Gaubert 2020](#); [Galle et al. 2023](#)). The distributional results also relate to

Finally, the paper relates to the broader quantitative literature on the welfare gains from trade and on the allocation of production and development. On the one hand, it develops a sufficient-statistics approach to measuring the gains from trade, following Arkolakis et al. (2012) and subsequent work (Costinot and Rodríguez-Clare 2014; Melitz and Redding 2015; Allen et al. 2020). On the other hand, it connects to the literature on the spatial allocation of production factors and macroeconomic development (see Desmet and Rossi-Hansberg 2014; Desmet et al. 2018). Relative to the first strand, endogenous firm location generates a new channel for the gains from trade, changes in the industrial base, that is quantitatively as important as the standard ACR term. Relative to the second, the framework endogenizes local technology through the equilibrium allocation of production, linking firm location to a central determinant of long-run development.

2 A Unified Firm Location Problem

This section develops a unified framework for endogenous firm location. The framework builds on Fajgelbaum et al. (2019) and generalizes it in two directions: it allows productivity draws to follow multivariate distributions with dependent components, and it specifies a location problem that can be embedded in several canonical market structures. As shown below, the dependence structure governs how firm allocation shapes the conditional productivity distribution in each location, which is the key object behind the industrial-base term in the welfare analysis.

To keep the results general, I need to impose a number of aggregate and firm-level restrictions. Consider a world economy composed of a finite set of countries $k \in \mathcal{K}$. There is a continuum of tradable goods indexed by ω . Labor is immobile across countries and is supplied exogenously, with $L_k > 0$ denoting the labor endowment of country k . Production in each country uses labor, with wage $w_k > 0$. Trade between countries is subject to bilateral iceberg trade costs $\tau_{kj} \geq 1$ for goods shipped from k to j as well as an ad valorem import tariff κ_{kj} imposed by destination country j .

Restriction 1. *In each destination j , the representative consumer has CES preferences over varieties with elasticity $\sigma > 1$.*

The endogenous firm location problem considered in this paper applies to models

work with multivariate Fréchet heterogeneity (Lind and Ramondo 2023a,b) and multivariate Pareto heterogeneity (Arkolakis et al. 2018, 2025).

that impose CES preferences as a macro-level restriction, similar to [Arkolakis et al. \(2012\)](#). In each destination j , the representative consumer has preferences over the set of available varieties Ω_j given by

$$U_j = \left[\int_{\omega \in \Omega_j} q_j(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1.$$

The associated CES price index is

$$P_j = \left[\int_{\omega \in \Omega_j} p_j(\omega)^{1-\sigma} d\omega \right]^{\frac{1}{1-\sigma}}. \quad (1)$$

As I show later, this restriction implies that bilateral sales are log-separable in exporter unit costs, bilateral barriers, and destination-market characteristics, which is central for the analytical tractability of the firm location problem.

Firms are indexed by productivity vectors $\mathbf{z}$ and can locate in any country $k \in \mathcal{K}$. Their productivities, however, need not be the same across locations and may depend on local conditions. I allow this dependence to be flexible and productivity draws across locations to range from independent to perfectly dependent. This is modeled using a multivariate distribution for the productivity vector $\mathbf{z} = (z_1, \dots, z_K)$, which leads to the next restriction.

Restriction 2. *The productivity vector $\mathbf{z} = (z_1, \dots, z_K)$ is distributed multivariate Fréchet with logistic dependence.*

The second restriction specifies the distribution of firm latent productivities across locations. While [Eaton and Kortum \(2002\)](#) use a special case of $F(\mathbf{z})$ under independence, they also discuss this cumulative distribution function with a general dependence structure. This distribution is also used in [Lind and Ramondo \(2023a,b\)](#) to examine correlated technologies. The results in this paper are complementary, but distinct: they imply that the productivity distribution of firms operating in a given country is not an exogenous primitive, but an equilibrium outcome shaped by firms' location choices. Hence, the mass of active firms and their productivity distribution are endogenous and jointly determined in equilibrium. This contrasts with standard approaches, which typically treat firm allocation and within-country heterogeneity as fixed primitives. Restriction 2 is not unique; I discuss alternative distributions in

Section 6. I discuss properties of this distribution in Appendix A.

I assume that $\mathbf{z}$ follows a multivariate Fréchet distribution with logistic dependence:

$$F(\mathbf{z}) = \exp \left(- \left(\sum_{k=1}^K [H_k z_k^{-\theta}]^\rho \right)^{1/\rho} \right), \quad H_k > 0, \quad \theta > 0, \quad \rho \geq 1. \quad (2)$$

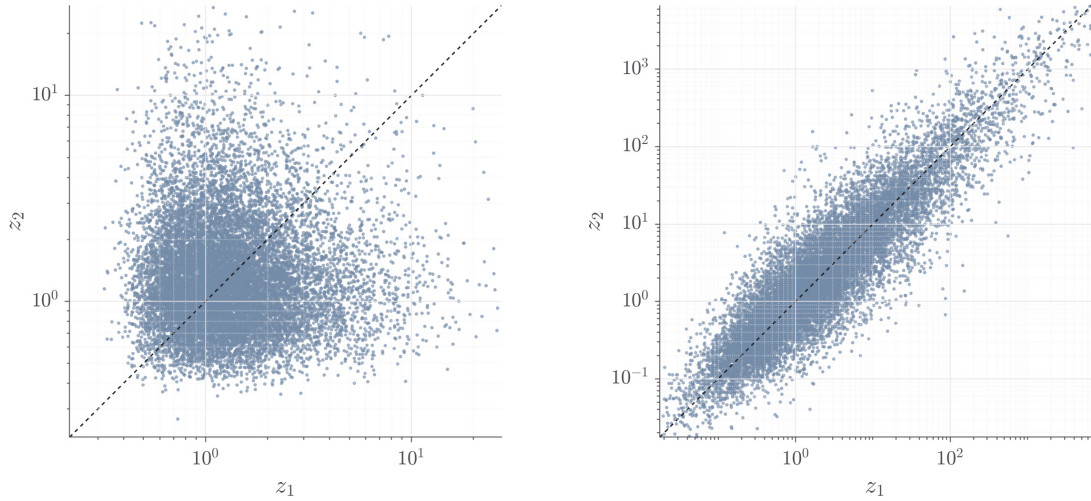
This distributional assumption guarantees several attractive properties. First, the unconditional marginal distribution of z_i is Fréchet with scale H_i and shape θ ,

$$F_i(z) = \exp \left(-H_i z^{-\theta} \right),$$

so that the interpretation of H_i and θ is consistent with [Eaton and Kortum \(2002\)](#). The parameter H_i is a country- i primitive that shifts the distribution of potential productivity draws for firms locating in i . The parameter θ governs the dispersion of productivity draws within countries.

(a) PANEL A: $\rho = 1$

(b) PANEL B: $\rho = 3$



Notes: The figure illustrates 15,000 simulated draws from a bivariate Fréchet distribution with common scale parameters $H_k = 1$ and tail parameter $\theta = 2$. Panel A shows the independence case, $\rho = 1$; Panel B shows the case with positive dependence, $\rho = 3$. The dashed line is the 45° diagonal. Both panels are plotted on a log-log scale.

Figure 1: SIMULATED DRAWS FROM BIVARIATE FRÉCHET DISTRIBUTION WITH LOGISTIC DEPENDENCE

The parameter ρ governs the dependence structure across locations: $\rho = 1$ corre-

sponds to independent productivity draws, while larger values of ρ allow for stronger positive dependence. In this sense, ρ provides a parsimonious measure of firms' potential mobility across countries. When $\rho = 1$, a high realization of z_i in country i carries no information about the firm's productivity in other countries. Location-specific productivity is then relatively important for the location decision. In the limit as $\rho \rightarrow \infty$, productivity is perfectly dependent across locations, after adjusting for country-specific scale parameters, and the location problem approaches the deterministic case. This distinction is illustrated in Figure 1. Panel A shows independent productivity draws, while Panel B shows positively dependent draws in the bivariate case.

The third restriction is at the firm level and requires the profit-to-revenue ratio to be constant across firms. The framework can accommodate different market structures as long as they satisfy this condition. Let $\mathcal{S}$ denote the profit-to-revenue ratio, and let $\mathcal{M}_i$ denote the market-access term faced by firms located in i . The class of market structures considered in this section is summarized by the following restriction.

Restriction 3. *A market structure is a tuple $(\eta, \mathcal{S}, \{\mathcal{M}_i\})$ with η and $\mathcal{S}$ invariant across locations.*

In addition to productivity draws in $\mathbf{z}$, firms take wages w_i and market access $\mathcal{M}_i$ as given when choosing its location. Total revenue from locating in i satisfies:

$$R_i(\mathbf{z}) \propto \left(\frac{z_i}{w_i} \right)^\eta \mathcal{M}_i, \quad \eta > 0, \quad (3)$$

where η is the elasticity of revenue with respect to cost-adjusted productivity z_i/w_i , and $\mathcal{M}_i$ is the market-access term. Total profits from locating in i are a constant fraction of revenue:

$$\Pi_i(\mathbf{z}) = \mathcal{S} R_i(\mathbf{z}). \quad (4)$$

Finally, I assume that firms' location decisions are equivalent to maximizing a strictly increasing transformation of total profits, or equivalently revenue, in this setting:

$$i^*(\mathbf{z}) = \arg \max_i \left\{ z_i w_i^{-1} \mathcal{M}_i^{1/\eta} \right\}. \quad (5)$$

Restrictions 1–3, together with the profit-maximization problem in equation (5), imply that firms choose locations based on productivity draws adjusted for production

costs and destination sales opportunities summarized by market access. The next proposition uses this structure to characterize firm allocation across discrete alternatives and the conditional productivity distribution among firms selected into each location.

Proposition 1. *Suppose the vector of shocks $\mathbf{z} = (z_1, \dots, z_K)$ has the joint cumulative distribution function F as in equation (2) and that agents choose the alternative k that maximizes the payoff in equation (5), then:*

(i) *The choice probabilities are given by:*

$$\mu_i = \frac{H_i^\rho w_i^{-\theta\rho} \mathcal{M}_i^{\theta\rho/\eta}}{\sum_k H_k^\rho w_k^{-\theta\rho} \mathcal{M}_k^{\theta\rho/\eta}}.$$

(ii) *The conditional distribution function of z_k given selection ($k^* = k$) is Fréchet:*

$$F_k(z \mid k^* = k) = \exp\left(-\tilde{T}_k z^{-\theta}\right) \quad , \text{ where } \quad \tilde{T}_k = H_k \mu_k^{-1/\rho}.$$

(iii) *If $\theta > 1$, the expected shock conditional on choosing k satisfies:*

$$\mathbb{E}[z_k \mid k^* = k] = H_k^{1/\theta} \mu_k^{-1/(\theta\rho)} \Gamma(1 - 1/\theta),$$

where $\Gamma(\cdot)$ denotes the Gamma function.

Proof. The proof is available in Appendix A. □

Proposition 1 shows how Fréchet heterogeneity within each country can arise endogenously from the firm location problem. Under a multivariate Fréchet distribution with logistic dependence, equilibrium location shares determine the scale of the conditional productivity distribution among firms that select into each country. In particular, the Fréchet scale in country k , $\tilde{T}_k = H_k \mu_k^{-1/\rho}$, combines the primitive technology term H_k with the endogenous selection term $\mu_k^{-1/\rho}$. This reinterpretation preserves the closed-form tractability of Fréchet-based trade models while endogenizing the allocation of firms, μ_k , across countries. Consequently, changes in fundamentals affect both the productivity distribution within countries and the equilibrium allocation of firms across locations.

Let T_k denote the industrial base of country k , interpreted as the supply shifter that

governs country k 's ability to serve every destination. It consists of the mass of firms that choose to locate in k with the productivity distribution of those firms. Using the conditional productivity scale in Proposition 1(iii), this object can be written as

$$T_k = \underbrace{\mu_k}_{\text{mass of firms}} \times \underbrace{\left(H_k^{1/\theta} \mu_k^{-1/(\theta\rho)} \right)^\eta}_{\text{conditional revenue productivity}}. \quad (6)$$

The first term captures the measure of producers located in country k , and therefore the measure of varieties which k can supply to the world economy. The second term captures the productivity of the firms that select into k . Productivity enters with exponent η because this is the elasticity that maps firm-level productivity advantages into revenue and aggregate supply capacity and depends on the market structure. This term can be viewed as the conditional aggregate revenue productivity.

The expression for the industrial base in equation (6) shows how the mass-composition tradeoff depends on the market-structure parameter η and on the dependence parameter ρ . It also clarifies the special cases and the role of the generalization developed in this paper. First, under perfect competition and Bertrand competition with independent productivity draws, $\eta = \theta$ and $\rho = 1$. In this case, the gain from attracting more firms is exactly offset by the lower conditional productivity of selected firms, so $T_k = H_k$. The location margin then disappears in the aggregate, and the framework collapses to Eaton and Kortum (2002) and Bernard et al. (2003). Second, under monopolistic competition, $\eta = \sigma - 1$ as in Suárez Serrato and Zidar (2016) and Fajgelbaum et al. (2019) and the dependence parameter ρ governs how strongly selection affects the industrial base. The general framework makes it possible to estimate both elasticity parameters from the data and to quantify the role of the industrial base in shaping the gains from trade.

3 Market Structures

I now embed the unified firm-location problem in three market structures: perfect competition, Bertrand competition, and monopolistic competition. The purpose of this section is to show that these environments share the same location-choice structure once three market-structure-specific objects are identified: the revenue elasticity η , the market-access aggregator $\mathcal{M}_i$, and the profit-to-sales ratio $\mathcal{S}$. Given these objects, Proposition 1 immediately delivers the equilibrium location shares and the

conditional productivity distributions of firms located in each country.

3.1 Perfect Competition and Bertrand Competition

I first embed the firm-location problem in two canonical models of international trade with Fréchet heterogeneity: the perfect-competition environment of [Eaton and Kortum \(2002\)](#) and the Bertrand-competition environment of [Bernard et al. \(2003\)](#). In both environments, goods are supplied by the lowest-cost producer, so aggregate trade outcomes are governed by country-level Fréchet productivity distributions. The key departure here is that these distributions are equilibrium outcomes rather than primitives. Conditional on them, the remaining derivations are standard and closely follow the canonical models.

Firms first choose production locations, and the firms that select into each country generate the productivity distribution from which its suppliers are drawn. This preserves the analytical structure of the canonical models while making the allocation of firms across countries endogenous. Given the revenue elasticity η , the profit-to-sales ratio $\mathcal{S}$, and the market-access term $\mathcal{M}_i$, [Proposition 1](#) directly delivers the equilibrium location shares and conditional productivity distributions.

To apply [Proposition 1](#), I give the conditional Fréchet distributions a firm-level interpretation. In the spirit of [Klette and Kortum \(2004\)](#), I treat each firm as owning a technology process that can be applied across the variety space. A firm located in country i generates cost offers for many varieties and supplies those for which it is the lowest-cost producer. The location-specific draw z_i therefore scales the firm's productivity process across goods rather than representing productivity for a single variety.

This firm-level interpretation is useful as it allows to connect directly to the results in [Proposition 1](#). Firm-level location decisions remain tractable and, conditional on the induced allocation of firms across country, aggregate results are consistent with [Eaton and Kortum \(2002\)](#) and [Bernard et al. \(2003\)](#). Conditional on locating in i , a firm with draw z_i generates a Fréchet efficiency distribution across varieties:

$$\Pr(\tilde{z}_i(\omega) \leq u \mid z_i) = \exp(-z_i^\theta u^{-\theta}).$$

Equivalently, the measure of varieties for which the firm's efficiency exceeds u is $z_i^\theta u^{-\theta}$. A higher z_i shifts the entire distribution of efficiencies across varieties: the measure of

varieties for which the firm draws high efficiency, and hence low delivered cost, rises with elasticity θ .

Given iceberg trade costs from i to j , τ_{ij} , and gross ad valorem tariff, κ_{ij} , a firm with location capability z_i that produces variety ω is able to deliver it to destination j at total unit cost:

$$c_{ij}(\omega) = \frac{w_i \tau_{ij} \kappa_{ij}}{\tilde{z}_i(\omega)}. \quad (7)$$

Consumers in destination j purchase each variety ω from the lowest-cost supplier. Under perfect competition, the lowest delivered unit cost is the consumer price. Under Bertrand competition, the lowest-cost producer charges a markup determined by the second-lowest cost, but this pricing rule does not affect the identity of the supplier. Hence, in both environments, trade shares and the measure of varieties won are governed by the same underlying cost race.

Since the firm's efficiency distribution across varieties is Fréchet with technology parameter z_i^θ , the measure of varieties that the firm can deliver to destination j at unit cost no greater than c is $z_i^\theta (w_i \tau_{ij} \kappa_{ij})^{-\theta} c^\theta$. Following standard steps, one can show that the distribution of the lowest delivered cost in destination j is governed by the aggregate cost parameter:

$$\Phi_j = \sum_k T_k (w_k \tau_{kj} \kappa_{kj})^{-\theta}, \quad (8)$$

where the effective scale T_k is determined by two objects: the mass of firms that select into country k and the conditional productivity distribution of those firms as defined in equation (6).

Let E_j denote total expenditure in destination j . Firms take aggregate objects Φ_j and E_j as given at the individual firm's location stage and total revenue of firm z is simply it's share in country j 's expenditure or:

$$R_{ij}(z_i) = z_i^\theta (w_i \tau_{ij})^{-\theta} \kappa_{ij}^{-(\theta+1)} \frac{E_j}{\Phi_j}. \quad (9)$$

Two comments are in order. First, firm-level revenue is proportional to the revenue shifter z_i^θ , which has a Pareto tail with index one, and hence does not have a finite mean. This creates no aggregation problem. Under perfect and Bertrand competi-

tion, destination expenditure is allocated through normalized sourcing probabilities generated by firms' offer intensities, rather than through the mean of z_i^θ . Second, a finite-firm approximation provides a useful way to see why the absence of a finite mean does not disturb the aggregate trade representation. The appropriate scaling is analogous to the normalization that arises for Zipf-tailed firm sizes from [Gabaix \(2011\)](#).⁵

Summing producer revenue over destinations gives the firm's total revenue and country i 's market access term:

$$R_i(z_i) = \sum_j R_{ij}(z_i) = \left(\frac{z_i}{w_i}\right)^\theta \mathcal{M}_i, \quad \mathcal{M}_i \equiv \sum_j \tau_{ij}^{-\theta} \kappa_{ij}^{-(\theta+1)} \frac{E_j}{\Phi_j}. \quad (10)$$

The profit expression is market-structure specific, but in both environments profits are a constant fraction of producer revenue $\Pi_i(z_i) = \mathcal{S} R_i(z_i)$, where $\mathcal{S}$ is common across origins, destinations, and firm types. Perfect competition and Bertrand competition differ in the constants governing markups, revenues, and profits, but not in the elasticity that maps firm productivity into sales or in the location statistic. In both environments, $\eta = \theta$. The two cases therefore imply the same firm-location problem up to the market-structure-specific constant $\mathcal{S}$.

Under perfect competition, operating profits are zero and $\mathcal{S} = 0$. If the location objective is interpreted literally as profit maximization, the zero-profit case is degenerate because all locations yield zero profits. There are two alternatives to define firm-location problem in Eaton-Kortum-type models. The first is to assume that

⁵To illustrate a finite-firm approximation, consider N potential firms, indexed by f . Let $N_i \equiv \sum_{f=1}^N \mathbf{1}\{k_f^* = i\}$ denote the realized number of firms selecting country i such that $(N_i/N) \xrightarrow{p} \mu_i$. Define the location-specific revenue shifter as $x_{fi} \equiv \mathbf{1}\{k_f^* = i\} z_{fi}^\theta$ and apply the results from Proposition 1 to get:

$$\Pr(x_{fi} > y) \sim \frac{\mu_i \tilde{T}_i}{y}.$$

Thus, the firm-level revenue shifter has the same tail index as firm size in the Zipf case considered by [Gabaix \(2011\)](#). The corresponding index-one stable-limit argument motivates scaling the source-specific sum by $N \ln N$, yielding:

$$\frac{1}{N \ln N} \sum_{f=1}^N x_{fi} \xrightarrow{p} \mu_i \tilde{T}_i = H_i \mu_i^{1-1/\rho} \equiv T_i,$$

where T_i is the industrial base term entering aggregate trade share. Since the $N \ln N$ normalization is common across firms and locations, it cancels from sourcing probabilities and country trade shares and does not affect firms' location choices and is therefore suppressed.

firms choose locations to maximize total producer revenue. The second is to view perfect competition as the limit of an environment with positive operating profits as $\mathcal{S} \rightarrow 0$. Under either interpretation, the firm-location problem is well defined and delivers the same allocation of firms across countries.

Bertrand competition leaves the fundamental cost comparison mechanism, and hence $\eta = \theta$ and $\mathcal{M}_i$, unchanged. What differs is the markup and therefore the profit-to-sales ratio. Let $c_{(1)j}$ and $c_{(2)j}$ denote the lowest and second-lowest delivered costs for a variety offered in j . Under Bertrand competition the lowest-cost supplier serves the variety at the limit price:

$$p_j = \min\{c_{(2)j}, \bar{m} c_{(1)j}\}, \quad \bar{m} = \frac{\sigma}{\sigma - 1},$$

The markup over the winner's marginal cost is:

$$m_j \equiv \frac{p_j}{c_{(1)j}} = \min \left\{ \frac{c_{(2)j}}{c_{(1)j}}, \bar{m} \right\}.$$

The joint distribution of $(c_{(1)j}, c_{(2)j})$ is the same for every variety and is independent of the identity of the lowest-cost supplier. Following [Bernard et al. \(2003\)](#), the unconstrained markup $c_{(2)j}/c_{(1)j}$ is Pareto with shape parameter θ . With the markup cap $\bar{m}$, the realized markup m_j has distribution:

$$\Pr(m_j \leq x) = 1 - x^{-\theta} \quad \text{for } 1 \leq x < \bar{m}, \quad \Pr(m_j = \bar{m}) = \bar{m}^{-\theta}.$$

In this environment profits are a constant fraction of producer revenue:

$$\mathcal{S} = \frac{1}{1 + \theta}.$$

Hence, Bertrand competition changes the level of operating profits relative to perfect competition, but because the profit-to-revenue ratio is common across locations, it does not change relative location payoffs conditional on the same revenue elasticity and market-access term.

Given the equilibrium effective Fréchet scales, T_i , defined in in equation (6) and using the result that $\eta = \theta$, international trade shares are characterized in the conventional

form:

$$\pi_{ij} = \frac{T_i (w_i \tau_{ij} \kappa_{ij})^{-\theta}}{\sum_k T_k (w_k \tau_{kj} \kappa_{kj})^{-\theta}} = \frac{H_i \mu_i^{1-1/\rho} (w_i \tau_{ij} \kappa_{ij})^{-\theta}}{\sum_k H_k \mu_k^{1-1/\rho} (w_k \tau_{kj} \kappa_{kj})^{-\theta}}. \quad (11)$$

This expression shows why allowing for a multivariate distribution of location-specific productivities is important. The dependence parameter ρ governs how the mass of firms and their conditional productivity composition interact. When $\rho = 1$, productivity draws are independent across locations, and the gain from attracting more firms is exactly offset by lower average productivity among selected firms. In this knife-edge case, $T_i = H_i$, so the location margin disappears and the model collapses to the standard Eaton–Kortum-type structures with fixed country-level productivity distributions. When $\rho > 1$, productivities are positively dependent across locations, $T_i = H_i \mu_i^{1-1/\rho}$ varies with the equilibrium allocation of firms. The industrial-base channel therefore becomes active: countries that attract more firms become more competitive suppliers in world markets.

The CES price index is as follows:

$$P_j = \mathcal{B}(\theta, \sigma) \Phi_j^{-1/\theta}, \quad \Phi_j = \sum_k H_k \mu_k^{1-1/\rho} (w_k \tau_{kj} \kappa_{kj})^{-\theta}, \quad (12)$$

where $\mathcal{B}(\theta, \sigma)$ is a constant that depends on the market structure but not on country-level fundamentals.

3.2 Monopolistic Competition

I next embed endogenous firm location in a monopolistic-competition environment. Unlike the perfect- and Bertrand-competition cases, where each firm owns a technology process spanning the variety space, each firm here produces a single differentiated variety. The environment is close in spirit to [Melitz \(2003\)](#), but I restrict attention to specifications in which export participation can be summarized by the same market-access term. This includes both the case without fixed export costs and the isomorphic case with stochastic fixed export costs proportional to destination-market size. Under these restrictions, fixed export costs affect the level of market access but not the structure of the location problem, and can therefore be suppressed.

Under [Restriction 1](#), a firm located in i with productivity z_i that serves destination

j charges the consumer price according to the standard constant mark-up rule:

$$p_{ij}(z_i) = \frac{\sigma}{\sigma - 1} \frac{w_i \tau_{ij} \kappa_{ij}}{z_i}.$$

Producer revenue, excluding tariff payments, is as follows:

$$R_{ij}(z_i) = z_i^{\sigma-1} \left(\frac{\sigma}{\sigma - 1} w_i \tau_{ij} \right)^{1-\sigma} \kappa_{ij}^{-\sigma} P_j^{\sigma-1} E_j. \quad (13)$$

Under monopolistic competition with CES demand, operating profit is a constant fraction of producer revenue:

$$\Pi_{ij}(z_i) = \frac{1}{\sigma} R_{ij}(z_i).$$

Thus, total operating profit of firm z conditional on locating in i is:

$$\Pi_i(z_i) = \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} \left(\frac{z_i}{w_i} \right)^{\sigma-1} \mathcal{M}_i, \quad (14)$$

where the market-access term is:

$$\mathcal{M}_i \equiv \sum_j \tau_{ij}^{1-\sigma} \kappa_{ij}^{-\sigma} \frac{E_j}{P_j^{1-\sigma}}. \quad (15)$$

Hence, monopolistic competition is characterized by

$$\eta = \sigma - 1, \quad \mathcal{S} = \frac{1}{\sigma},$$

with market access given by equation (15). As before, the mass of firms located in i is μ_i , with the total mass of firms normalized to one. I can now apply the results in Proposition 1 such that conditional on selection into i , firm productivity is Fréchet with scale:

$$\tilde{T}_i = H_i \mu_i^{-1/\rho}.$$

Thus, $\tilde{T}_i$ is the per-firm productivity scale among firms located in i , while the aggregate source term also depends on the selected mass μ_i .

Integrating producer revenue over the conditional productivity distribution of firms

located in i , average bilateral producer revenue in destination j is

$$\int R_{ij}(z) dF_i(z \mid i^* = i) = \mathcal{C}(\sigma, \theta) \tilde{T}_i^{(\sigma-1)/\theta} w_i^{1-\sigma} \tau_{ij}^{1-\sigma} \kappa_{ij}^{-\sigma} P_j^{\sigma-1} E_j, \quad (16)$$

where $\mathcal{C}(\sigma, \theta)$ is a constant; as is standard in the literature, I assume that $\theta > \sigma - 1$.

Consumer expenditure on varieties produced in i is κ_{ij} times producer revenue. Hence, international trade shares can be written as:

$$\pi_{ij} = \frac{H_i^{(\sigma-1)/\theta} \mu_i^{1-(\sigma-1)/(\rho\theta)} (w_i \tau_{ij} \kappa_{ij})^{1-\sigma}}{\sum_k H_k^{(\sigma-1)/\theta} \mu_k^{1-(\sigma-1)/(\rho\theta)} (w_k \tau_{kj} \kappa_{kj})^{1-\sigma}}. \quad (17)$$

The CES price index satisfies:

$$P_j^{1-\sigma} = \sum_i \mu_i \int p_{ij}(z)^{1-\sigma} dF_i(z \mid i^* = i). \quad (18)$$

Using the conditional Fréchet distribution from Proposition 1, it can be written as follows:

$$P_j^{1-\sigma} = \mathcal{D}(\sigma, \theta) \sum_i H_i^{(\sigma-1)/\theta} \mu_i^{1-(\sigma-1)/(\rho\theta)} (w_i \tau_{ij} \kappa_{ij})^{1-\sigma}, \quad (19)$$

where $\mathcal{D}(\sigma, \theta)$ is a constant.

4 Aggregate Mappings and Equilibrium

The previous section shows that the endogenous firm-location problem can be embedded in perfect competition, Bertrand competition, and monopolistic competition in a unified way. Although these market structures differ in their microeconomic foundations, each is summarized by the same aggregate objects: a revenue elasticity η , a profit-to-revenue ratio $\mathcal{S}$, and a market-access term $\mathcal{M}_i$. Once these objects are specified, location shares, trade shares, price indices, and welfare can be written in an isomorphic form. This section collects these aggregate mappings and defines the equilibrium system conditional on the market-structure parameters.

Given the allocation of firms in Proposition 1, destination j 's consumer-price expen-

diture share on goods sourced from i is

$$\pi_{ij} = \frac{H_i^{\eta/\theta} \mu_i^{1-\eta/(\theta\rho)} (w_i \tau_{ij} \kappa_{ij})^{-\eta}}{\sum_k H_k^{\eta/\theta} \mu_k^{1-\eta/(\theta\rho)} (w_k \tau_{kj} \kappa_{kj})^{-\eta}}. \quad (20)$$

The market-access term relevant for the location decision is

$$\mathcal{M}_i = \sum_j \frac{\tau_{ij}^{-\eta} \kappa_{ij}^{-(\eta+1)}}{\sum_k H_k^{\eta/\theta} \mu_k^{1-\eta/(\theta\rho)} (w_k \tau_{kj} \kappa_{kj})^{-\eta}} E_j. \quad (21)$$

Finally, the aggregate CES price index satisfies

$$P_j^{-\eta} = \mathcal{A} \sum_i H_i^{\eta/\theta} \mu_i^{1-\eta/(\theta\rho)} (w_i \tau_{ij} \kappa_{ij})^{-\eta}, \quad (22)$$

where $\mathcal{A}$ is a market-structure constant.

Given the aggregate variables $\{\mu_i, \pi_{ij}, \mathcal{M}_i, P_i\}$, it remains to specify aggregate expenditure and the equilibrium condition that pins down wages. Aggregate expenditure in country i consists of labor income, redistributed profits, tariff revenue, and an exogenous trade deficit D_i :

$$E_i = w_i L_i + \Pi_i + TR_i + D_i. \quad (23)$$

Since E_j is final expenditure at consumer prices while producers receive revenue net of tariffs, the producer-price trade flow from i to j is as follows:

$$X_{ij} = \frac{\pi_{ij}}{\kappa_{ij}} E_j. \quad (24)$$

Tariff revenue collected by destination j is therefore:

$$TR_j = \sum_i (\kappa_{ij} - 1) X_{ij} = (1 - \Lambda_j) E_j, \quad (25)$$

where the wedge Λ_j is the share of destination j 's tariff-inclusive expenditure that is received by producers at producer prices:

$$\Lambda_j = \sum_i \frac{\pi_{ij}}{\kappa_{ij}}. \quad (26)$$

Equivalently, $1 - \Lambda_j$ is the share of expenditure collected as tariff revenue. When $\kappa_{ij} = 1$ for all sources, $\Lambda_j = 1$ and tariff revenue is zero.

Producer revenue in country i is the sum of producer-price sales across destinations:

$$Y_i = \sum_j X_{ij} = w_i L_i + \Pi_i.$$

This holds because in equilibrium producer revenue is distributed to domestic labor as factor income and profits. Combining this identity with the budget constraint and the expression for tariff revenue gives:

$$E_i = Y_i + (1 - \Lambda_i)E_i + D_i.$$

Thus the equilibrium trade-balance condition is:

$$\Lambda_i E_i = Y_i + D_i. \tag{27}$$

This allows me to define equilibrium.

Definition 1. For a fixed market structure $(\eta, \mathcal{S})$, technology parameters $\{H_i\}$, common parameters (θ, ρ) , labor endowments $\{L_i\}$, trade costs $\{\tau_{ij}\}$, tariffs $\{\kappa_{ij}\}$, and deficit parameters $\{d_i\}$, an equilibrium is a vector of wages $\{w_i\}$, final expenditures $\{E_i\}$, producer revenues $\{Y_i\}$, firm-location shares $\{\mu_i\}$, market-access terms $\{\mathcal{M}_i\}$, trade shares $\{\pi_{ij}\}$, and price indices $\{P_i\}$ such that:

- (i) *Trade and prices.* Trade shares, market access, and price indices satisfy (20), (21), and (22).
- (ii) *Location.* Firm-location shares satisfy the expression in Proposition 1.
- (iii) *Income and expenditure.* Producer revenues and final expenditures satisfy (23)-(27).

5 Welfare

This section derives welfare implications of the unified representation using sufficient statistics. First, as in the standard gains-from-trade formula of Arkolakis et al. (2012), welfare changes depend on changes in the domestic expenditure share and the trade

elasticity. Second, as in [Demidova and Rodríguez-Clare \(2009\)](#), [Felbermayr et al. \(2013\)](#), and [Lashkaripour \(2021\)](#), tariff revenues will also shape welfare. Most importantly, welfare depends on an additional sufficient statistic: the change in country i 's industrial base.

I measure welfare in country i by real final expenditure per worker defined as:

$$W_i \equiv \frac{E_i}{P_i L_i}.$$

I assume throughout that the trade deficit is fixed relative to producer income according to the following country-specific constants:

$$d_i = \frac{D_i}{Y_i}.$$

For a discussion of how this assumption affects welfare estimates, see [Costinot and Rodríguez-Clare \(2014\)](#). Using the trade-balance condition in [\(27\)](#) and the factor-income identity $w_i L_i = (1 - \mathcal{S})Y_i$, welfare can be written as follows:

$$W_i = \underbrace{\frac{w_i}{P_i}}_{\text{real wage}} \times \underbrace{\frac{1}{\Lambda_i}}_{\text{tariff wedge}} \times \underbrace{\frac{1 + d_i}{1 - \mathcal{S}}}_{\text{country constant}}. \quad (28)$$

Thus welfare depends on the real wage w_i/P_i , the tariff-revenue wedge Λ_i , and country-specific constant term that consists of the deficit-to-income ratio d_i and the profit share $\mathcal{S}$. Together with the expression for the intra-trade shares, this representation leads to the following proposition that derives sufficient-statistic welfare formula.

Proposition 2. *Let $\Delta \ln x_i \equiv \ln x'_i - \ln x_i$ denote the log change between an initial equilibrium and a counterfactual equilibrium. Holding H_i , d_i , and the market structure fixed, the change in welfare is as follows:*

$$\Delta \ln W_i = -\frac{1}{\eta} \Delta \ln \pi_{ii} + \left(\frac{1}{\eta} - \frac{1}{\theta \rho} \right) \Delta \ln \mu_i - \Delta \ln \Lambda_i. \quad (29)$$

Proof. See Appendix B. □

Equation [\(29\)](#) isolates three relevant welfare channels. The first term $-\frac{1}{\eta} \Delta \ln \pi_{ii}$ is

the ACR statistics with the trade elasticity η determined by the market structure: $\eta = \theta$ under cost competition and $\eta = \sigma - 1$ under monopolistic competition. The second term is the industrial-base term, which is the novel channel that appears due to the endogenous-location mechanism. Finally, the term $-\Delta \ln \Lambda_i$ is the tariff-revenue channel.

The structure of the model makes it possible to recast Proposition 2 in terms of an alternative, and arguably more readily observable, sufficient statistic. Let $\nu_i \equiv Y_i / \sum_k Y_k$ denote country i 's share of world producer revenue. In equilibrium $\mu_i = \nu_i$, the welfare change can be written as follows:

$$\Delta \ln W_i = -\frac{1}{\eta} \Delta \ln \pi_{ii} + \left(\frac{1}{\eta} - \frac{1}{\theta \rho} \right) \Delta \ln \nu_i - \Delta \ln \Lambda_i. \quad (30)$$

To see the equivalence, sum (24) across destinations and use (21) to obtain $Y_i = H_i^{\eta/\theta} \mu_i^{1-\eta/(\theta\rho)} w_i^{-\eta} \mathcal{M}_i$ and combine with the identity of the location-share equation in Proposition 1 and the property that $\sum_i \mu_i = 1$.

Despite this convenience, I retain μ_i as the primitive object. First, the model's mechanisms are functions of the mass of firms, not of revenue: selection dilutes conditional productivity through $\mu_i^{-1/\rho}$, the industrial base is $T_i = H_i \mu_i^{1-1/\rho}$, and at $\rho = 1$ the location margin vanishes from aggregates while revenue shares do not. Second, the direction of use differs: ex post, ν_i is observable and (30) evaluates welfare from data, but counterfactuals solve for μ_i through the structural mapping, with ν_i determined as an outcome.

Next, I consider several special cases of the general formula in equation (29). In case of perfect and Bertrand competition, $\eta = \theta$, so (29) becomes:

$$\Delta \ln W_i = -\frac{1}{\theta} \Delta \ln \pi_{ii} + \frac{\rho - 1}{\theta \rho} \Delta \ln \mu_i - \Delta \ln \Lambda_i. \quad (31)$$

The location coefficient $\frac{\rho-1}{\theta\rho}$ vanishes when technology draws are independent across locations, $\rho = 1$. Moreover, under the assumption of balanced trade and no tariffs, (31) collapses to the standard ACR sufficient statistic.

Under monopolistic competition, $\eta = \sigma - 1$, and the coefficient on the location term

is:

$$\frac{1}{\eta} - \frac{1}{\theta\rho} = \frac{1}{\sigma - 1} - \frac{1}{\theta\rho}.$$

This coefficient is strictly positive at $\rho = 1$ under the standard restriction $\theta > \sigma - 1$. Hence, unlike in the perfect-competition and Bertrand cases, the endogenous-location channel remains active under independent draws in monopolistic competition. Its quantitative strength, however, depends on ρ and must therefore be disciplined empirically.

6 Alternative Distributions

The selection mechanism developed in this paper is not unique to the multivariate Fréchet distribution. The key requirement is that equilibrium location shares tractably determine the conditional productivity distribution of firms selecting into each location. The Fréchet specification is convenient because it preserves standard closed-form aggregation, but other distributions can generate the same selection logic.

I also consider an alternative distributional foundation that connects the framework to the literature on power-law heterogeneity (see, e.g., [Gabaix 2009](#)). Proposition 3 studies a multivariate Dagum distribution with Clayton dependence. This distribution delivers the same economic mechanism: firm location choices induce conditional productivity distributions, and the dependence structure governs how the mass and composition of firms interact in aggregate productivity. So, while the main quantitative analysis uses the Fréchet formulation because it nests directly in canonical quantitative trade models, the power-law case shows that the mechanism is not specific to extreme-value heterogeneity.

Proposition 3. *Suppose the vector of shocks $\mathbf{z} = (z_1, \dots, z_K)$ follows a Dagum distribution with Clayton-type dependence:*

$$F(\mathbf{z}) = \left(1 + \sum_{k=1}^K \left[\frac{z_k}{H_k} \right]^{-\xi} \right)^{-1/\delta}, \quad H_k > 0, \xi > 0, \delta > 0,$$

where parameter ξ governs the power-law tail of post-selection heterogeneity, while δ governs dependence across alternatives. Agents choose the alternative k that maximizes the payoff in equation (5). Then:

(i) The choice probabilities are given by:

$$\mu_i = \frac{H_i^\xi w_i^{-\xi} \mathcal{M}_i^{\xi/\eta}}{\sum_k H_k^\xi w_k^{-\xi} \mathcal{M}_k^{\xi/\eta}}.$$

(ii) The conditional distribution of z_k given selection ($k^* = k$) is Dagum:

$$F_k(z \mid k^* = k) = \left[1 + \left(\frac{z}{\tilde{D}_k} \right)^{-\xi} \right]^{-1/\delta}, \quad \text{where} \quad \tilde{D}_k = H_k \mu_k^{-1/\xi}.$$

(iii) If $\xi > 1$, the expected shock conditional on selection is

$$\mathbb{E}[z_k \mid k^* = k] = \frac{\Gamma(1 - 1/\xi) \Gamma(1/\delta + 1/\xi)}{\Gamma(1/\delta)} H_k \mu_k^{-1/\xi},$$

where $\Gamma(\cdot)$ denotes the Gamma function.

Proof. The proof is available in Appendix C. □

The Fréchet and Dagum specifications are tractable for the same economic reason. In both cases, the productivity vector can be interpreted as having two components:

$$z_k = \bar{Z} Z_k.$$

The common component $\bar{Z}$ raises or lowers productivity in all locations for a given firm, while the idiosyncratic component Z_k determines which location is relatively most attractive. Since $\bar{Z}$ is common across locations, it cancels from the location decision. The firm's choice is therefore governed by the relative draws Z_k , while $\bar{Z}$ determines how strongly productivity draws are correlated across locations.

The two distributions differ in how dependence affects relative draws. In the logistic-Fréchet case, stronger dependence changes the effective tail of the idiosyncratic component that ranks locations. The dependence parameter therefore enters the location shares directly. In the Clayton-Dagum case, dependence operates differently. It changes the joint distribution and the moments conditional on selection, but it does not change the tail parameter that governs relative choice probabilities. As a result, the dependence parameter appears in the conditional productivity distribution

only through terms that are common across alternatives. Taken together, the two cases cover the two main distributional foundations for firm heterogeneity used in the literature: extreme-value heterogeneity and power-law heterogeneity.

7 Estimation

This section quantifies the importance of the industrial-base channel generated by endogenous firm location in otherwise canonical models of international trade. Proposition 2 shows that, relative to standard sufficient-statistics approach based on bilateral trade flows X_{ij} , the model requires two additional inputs: measures of each country's industrial base, μ_i , and the structural parameters $\{\eta, \theta, \rho\}$.

I combine several data sources to structurally estimate the firm-location shares μ_i and the parameters $\{\eta, \theta, \rho\}$ that jointly determine the quantitative importance of the industrial-base channel. International trade flows and domestic production are from the International Trade and Production Database described in [Borchert et al. \(2021\)](#) and [Larch et al. \(2025\)](#). I focus on manufacturing trade and observe bilateral trade flows X_{ij} including domestic absorption. International tariff data are from Feodora Teti's Global Tariff Database described in [Teti \(2024\)](#). Bilateral distance is from the CEPII database described in [Mayer and Zignago \(2011\)](#). In estimation, I also use country-level controls from the Penn World Table v.11 described in [Feenstra et al. \(2015\)](#).

I proceed in two steps. The first step estimates the trade elasticity η from tariff variation in a structural gravity equation. This step also recovers exporter-year fixed effects, importer-year fixed effects, and the bilateral component of trade costs. The second step uses these recovered gravity components to construct market-access terms and estimate the composite parameter that governs the strength of the industrial-base channel in the welfare formula. I then use the model structure to recover the location shares μ_i .

While the model is static, the estimation uses panel variation in the data. Let t index years. The empirical counterpart of the producer-price trade flow in equation (24) is estimated by Poisson Pseudo-Maximum Likelihood:

$$X_{ij,t} = \exp(v_{i,t} + \log b_{ij,t} + \chi_{j,t}) + e_{ij,t}, \quad (32)$$

where $v_{i,t}$ is an exporter-year fixed effect, $\chi_{j,t}$ is an importer-year fixed effect, and $b_{ij,t}$ is the bilateral component that captures all trade barriers including tariffs. Combining the trade flow equation with the results in Proposition 1 allows to express the fixed effects as the following log-linear equations:

$$v_{i,t} = \rho \log H_{i,t} - \theta \rho \log w_{i,t} + \left(\frac{\theta \rho}{\eta} - 1 \right) \log \mathcal{M}_{i,t}, \quad \chi_{j,t} = \log E_{j,t} - \log \Phi_{j,t}.$$

For estimation, I parameterize bilateral trade barriers as:

$$\log b_{ij,t} = \psi_{ij} + \varphi_{q(ij)} t + \beta \log \kappa_{ij,t}, \quad (33)$$

where ψ_{ij} is a pair fixed effect, $q(ij)$ denotes the log-distance decile of pair ij such that $\varphi_{q(ij)} t$ allows for different trends across distance bins. Since in the data $X_{ij,t}$ is measured in CIF terms but net of tariffs, the model implies:

$$\beta = -(\eta + 1).$$

Thus, the trade elasticity parameter can be recovered as $\hat{\eta} = -\hat{\beta} - 1$ reported in Table 1. The tariff coefficient is -3.6 , implying $\hat{\eta} = 2.6$. This estimate sits between the two main strands of the literature: above the lower long-run trade elasticity of 1.75 – 2.25 estimated by [Boehm et al. \(2023\)](#), and below the value of 3.6 used in [Eaton and Kortum \(2002\)](#) and [Bernard et al. \(2003\)](#).

Table 1: FIRST-STAGE STRUCTURAL GRAVITY ESTIMATION

	Coefficient
$\log \kappa_{ij,t}$	-3.593 (0.285)
Implied $\hat{\eta}$	2.593
Exporter-year fixed effects	Yes
Importer-year fixed effects	Yes
Pair fixed effects	Yes
Distance-decile trends	Yes
Observations	802,086
Pair clusters	29,398

Notes: The regression is estimated by PPML and includes exporter-year, importer-year, and pair fixed effects as well as log-distance-decile-specific linear trends. Robust standard errors clustered by country pair are reported in parentheses.

In the next stage, I use the exporter-specific component recovered in the first stage and run the following panel model:

$$\widehat{v}_{i,t} = \text{country}_i + \text{year}_t + \gamma \log \widehat{\mathcal{M}}_{i,t} + \mathbf{W}'_{i,t} \Gamma + u_{i,t}, \quad (34)$$

where country_i and year_t are exporter and year fixed effects. The coefficient on $\log$ market access identifies $\gamma = \left(\frac{\theta\rho}{\eta} - 1\right)$. Note that under Bertrand and perfect competition, $\eta = \theta$ such that ρ can be inferred from $\gamma + 1$. The vector $\mathbf{W}_{i,t}$ denotes variables used to control for the time-varying components of $H_{i,t}$ and $w_{i,t}$: $\log$ GDP per capita, $\log$ capital per worker, the human capital index, the $\log$ output price level, the labor share, $\log$ TFP, and the $\log$ terms of trade all from the Penn World Table.

Estimating γ requires observations of market access. As derived by Fally (2015), estimating the gravity equation via PPML automatically satisfies the adding up constraints, which together with the theoretical identities of exporter and importer fixed effects allow to recover a direct measure of market access:

$$\widehat{\mathcal{M}}_{i,t} = \sum_j \widehat{b}_{ij,t} \exp(\widehat{\chi}_{j,t}).$$

I first estimate the panel model in equation (34) by OLS and report the results in Table 2. The coefficient on $\log \widehat{\mathcal{M}}_{i,t}$ is precisely estimated at 0.61, implying $\widehat{\rho} = 1.61$ under the assumption $\eta = \theta$. The estimate points to a substantial departure from the independent benchmark, $\rho = 1$. OLS estimates, however, may be biased for two reasons. First, $\widehat{\mathcal{M}}_{i,t}$ is a generated regressor, so measurement error may attenuate the coefficient toward zero. Second, contemporaneous shocks may jointly affect recovered market access, $\widehat{\mathcal{M}}_{i,t}$, and recovered exporter competitiveness, $\widehat{v}_{i,t}$. To address these concerns, I use two model-consistent instruments based on predicted variation in foreign market access that is plausibly orthogonal to contemporaneous exporter shocks. I also adjust inference to account for the fact that the dependent variable is generated in the first-stage estimation.

The first instrument fixes destination attractiveness at the start of the panel and relies on the time-series variation in the estimated bilateral component:

$$Z_{i,t}^{(I)} = \sum_{j \neq i} \exp\left(\widehat{\psi}_{ij} + \widehat{\varphi}_{q(ij)}(t - t_0) - (\widehat{\eta} + 1) \log(\kappa_{ij,t})\right) \exp(\widehat{\chi}_{j,t_0}).$$

The instrument excludes the domestic market and removes contemporaneous destination-demand movements from market access. Its time-series variation comes from the estimated evolution of bilateral trade costs, including distance-decile trends and tariff changes, weighted by destination attractiveness in the base year. The identifying assumption is that the resulting variation in $Z_{i,t}^{(I)}$ affects the recovered exporter-year component only through market access.

The second instrument is more conservative. It replaces actual bilateral tariffs with MFN tariffs and omits the estimated distance-decile trend:

$$Z_{i,t}^{(II)} = \sum_{j \neq i} \exp\left(\widehat{\psi}_{ij} - (\widehat{\eta} + 1) \log(\kappa_{ij,t}^{MFN})\right) \exp(\widehat{\chi}_{j,t_0}).$$

The intuition follows [Boehm et al. \(2023\)](#). While destination j may be able to target tariffs toward a specific partner i , changes in MFN rates apply to many partners and are therefore less likely to reflect bilateral shocks specific to i . The second instrument therefore uses only pair-specific exposure, base-year destination attractiveness, and MFN-tariff-predicted changes in foreign market access. This removes contemporaneous destination-demand variation, excludes the domestic market, and eliminates variation coming from preferential bilateral tariffs.

Table 2: SECOND-STAGE ESTIMATION

	OLS	IV (I)	IV (II)
$\log \widehat{\mathcal{M}}_{i,t}$	0.559	1.083	1.230
Clustered SE	(0.186)	(0.402)	(0.549)
Joint conditional bootstrap SE	[0.219]	[0.457]	[0.624]
Exporter fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	3,189	3,189	3,189
Exporter clusters	117	117	117
Kleibergen–Paap F	–	23.93	11.85

Notes: The dependent variable is the exporter-year fixed effect $\widehat{v}_{i,t}$ recovered from the first-step gravity regression. Column 1 reports the OLS coefficient. Column 2 instruments market access with predicted foreign access from the full estimated bilateral component and base-year 1992 importer effects; column 3 uses MFN-tariff-predicted foreign access and base-year importer effects. Controls include log GDP per capita, log capital per worker, human capital, the output price level, the labor share, TFP, and the terms of trade. Standard errors reported in parentheses are clustered by exporter. Joint conditional bootstrap standard errors based are reported in brackets.

The IV estimates in Table 2 exceed the OLS estimate, consistent with attenuation from potential measurement errors. Using $Z_{i,t}^{(I)}$, the coefficient on market access is 1.08, implying $\hat{\rho} = 2.08$ under $\eta = \theta$. Using the MFN-based instrument $Z_{i,t}^{(II)}$, the coefficient is 1.23, implying $\hat{\rho} = 2.23$. The weak-identification diagnostics are strong for $Z_{i,t}^{(I)}$, with a Kleibergen-Paap F-statistic of 23.9, and remain acceptable for the more conservative MFN-based instrument $Z_{i,t}^{(II)}$, with $F = 11.9$. Both IV estimates point to strong positive dependence in location-specific productivity draws.

I also account for the fact that the dependent variable in the second stage is generated in the first-stage gravity regression and implement a joint conditional bootstrap that combines first-stage sampling uncertainty in the generated $\hat{v}_{i,t}$ with second-stage exporter-level dependence, while holding the second-stage design fixed. In each replication, I resample bilateral pairs with replacement and re-estimate the first-stage PPML with the tariff coefficient fixed at its benchmark estimate, so that the resulting standard errors are conditional on $\hat{\eta}$. I then recover the bootstrapped exporter-year fixed effect, $\hat{v}_{i,t}^{G,b}$. Holding fixed the market-access measure, instruments, controls, sample, and fixed-effect structure, I combine this generated- $\hat{v}$ perturbation with a wild-cluster perturbation of the second-stage residual at the exporter level. Specifically, for specification s , the bootstrap dependent variable is

$$\hat{v}_{i,t}^{b,s} = \hat{v}_{i,t}^{G,b} + (\xi_i^b - 1) \hat{u}_{i,t}^s, \quad (35)$$

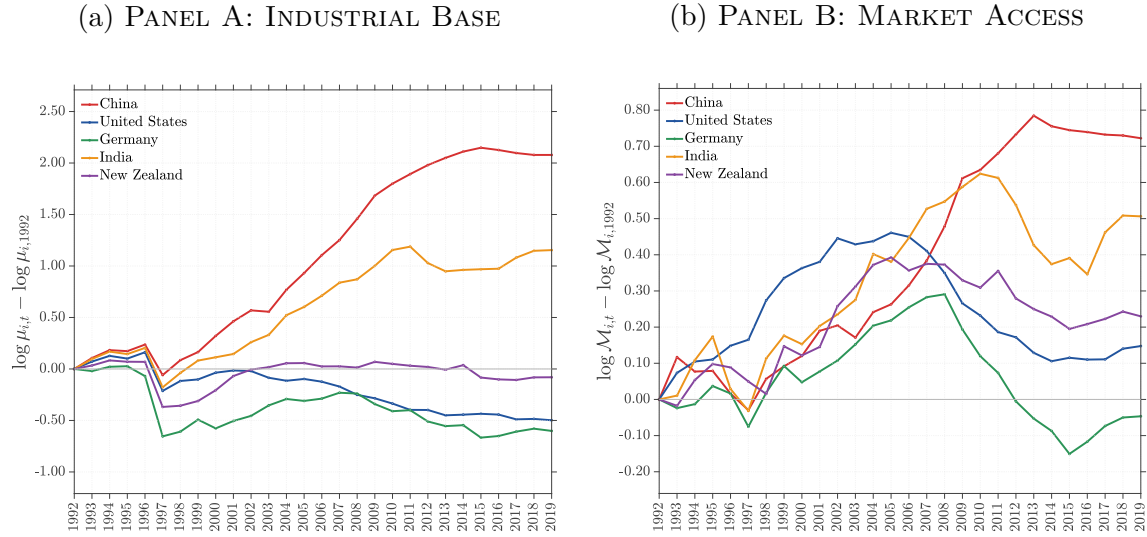
where ξ_i^b is an exporter-level Rademacher multiplier, equal to -1 or 1 with equal probability, and $\hat{u}_{i,t}^s$ is the benchmark second-stage residual from specification s , net of the exporter and year fixed effects. I then re-estimate the second-stage specification on the fixed right-hand-side variables and reported the resulting bootstrap standard errors in Table 2. As expected, the joint conditional bootstrap produces standard errors that are larger than the exporter-clustered standard errors. Overall, accounting jointly for the generated dependent variable and exporter-level second-stage uncertainty increases standard errors but leaves the estimated coefficients relatively precise.

Finally, in addition to the market access term $\mathcal{M}_i$ the estimation procedure also allows me to calculate the mass of producers μ_i that locate in country i . Together, these objects summarize the paper’s new mechanism: market access raises the value of producing in a country, while location selection determines the mass of firms operating there. The model-implied firm-location share can be recovered from the exporter fixed

effect and market access as:

$$\hat{\mu}_{i,t} = \frac{\exp(\hat{v}_{i,t})\hat{\mathcal{M}}_{i,t}}{\sum_{\ell} \exp(\hat{v}_{\ell,t})\hat{\mathcal{M}}_{\ell,t}}.$$

Figure 2 reports changes in the recovered mass of producers and market access for selected countries over 1992–2019 relative to the benchmark year of 1992. The figure decomposes the evolution of exporter competitiveness into the recovered firm-location share, in Panel A, and recovered market access, in Panel B. The figure highlights two broad patterns.



Notes: Each series is expressed as a log change relative to 1992. The left panel reports changes in the recovered firm-location share, $\ln \hat{\mu}_{i,t} - \ln \hat{\mu}_{i,1992}$. The right panel reports changes in recovered market access, $\ln \hat{\mathcal{M}}_{i,t} - \ln \hat{\mathcal{M}}_{i,1992}$.

Figure 2: RECOVERED CHANGES IN MASS OF PRODUCERS AND MARKET ACCESS

First, China and India both experience large increases in market access and firm-location shares, but the timing and persistence of the two margins differ. For China, recovered market access grows steadily after WTO accession in 2001 and peaks in the early 2010s, consistent with its integration into the global economy. The recovered location share rises even more sharply through the mid-2010s before flattening, suggesting that improved market access reallocated the world production base toward China. India also records sizable gains in both variables peaking around 2010-2011. Market access then falls sharply during the global trade slowdown of the mid-2010s before recovering, while the location share plateaus rather than tracking market ac-

cess downward. In both cases, the location margin is distinct from contemporaneous market access and appears more persistent.

Second, several developed economies display the two margins moving in opposite directions. Market access for the United States improves early, with gains concentrated in the 1990s, well before China’s rise. After 2008, market access reverses for both the United States and Germany, while their recovered location shares continue to decline consistent with a reallocation of production toward faster-growing and lower-cost locations. Being a relatively distant market, New Zealand is a useful contrast: its market access improves by roughly 0.25 log points, while its location share remains essentially flat, suggesting a destination-demand gain that does not attract additional production.

7.1 Internal Validation: A Model-Based Monte Carlo

The estimation procedure combines the structural gravity equation with the model-implied relationship between exporter competitiveness and market access. A natural concern is that the log-linear gravity equation in (32) is observationally similar across many trade models. Then the relevant question is whether an economy without location choice can mechanically generate a positive relationship between recovered exporter fixed effects and constructed market access. If it can, the estimated market-access coefficient may partly reflect a generic gravity correlation rather than the location-selection channel.

I assess this concern with a model-based Monte Carlo. The exercise asks whether the two-step procedure can theoretically recover the correct market-access relationship in data generated by the model with endogenous firm location choice, and whether it returns the null value in data generated by an otherwise identical economy without location choice. The exercise therefore provides an internal validation of the identification logic. Admittedly, it is not designed to distinguish the mechanism from alternatives outside the model; I provide external validation that does not depend on the specific model structure in the next section.

I simulate two world economies that share the same primitives and the same sequence of trade-cost shocks, which generate variation in trade flows across equilibria interpreted as time periods. The two versions differ only in whether the location-choice margin is active. I calibrate the simulation to the preferred empirical IV estimate,

$\gamma = 1.23$. In the no-location economy, trade-cost shocks affect trade flows and wages but do not reallocate firms across origins. This placebo economy therefore has the same gravity structure as the location-choice economy, but shuts down endogenous selection into production locations.

I simulate the world economy for $K = 40$ countries and $T = 25$ periods in each Monte Carlo replication, and repeat the exercise for 100 replications. For each period, I solve the static equilibrium implied by the current realization of trade costs and record bilateral trade flows. The researcher observes simulated flows $X_{ij,t}^{\text{obs}}$, trade costs $\tau_{ij,t}$, and wages $w_{i,t}$. The parameter η is treated as known. Further details on the Monte Carlo simulations are reported in Appendix D.

The simulated data sets are then passed through the same two-step estimation procedure used in the empirical analysis. Table 3 reports the results. In the location-choice economy, the target value is $\gamma = 1.230$, matching the preferred empirical IV estimate used to calibrate the Monte Carlo. The mean OLS estimate is 1.195 and the mean IV estimate is 1.224, both close to the target.

Table 3: MONTE CARLO VALIDATION OF ESTIMATION PROCEDURE

	Location choice		No location choice	
	OLS	IV	OLS	IV
Mean γ	1.195	1.224	0.003	-0.001
Std. Dev.	0.047	0.068	0.027	0.030
Target γ	1.230	1.230	0.000	0.000
Exporter fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Countries per replication	40	40	40	40
Periods per replication	25	25	25	25
Replications	100	100	100	100

Notes: Entries report means and standard deviations across 100 Monte Carlo replications. In the location-choice economy, the simulation is calibrated to the preferred empirical IV estimate, $\gamma = 1.23$. The no-location-choice economy uses the same primitives and trade-cost shocks but holds the source term fixed, shutting down endogenous reallocation of firms across origins.

In the no-location economy, the target is $\gamma = 0$. Both OLS and IV return coefficients essentially equal to zero: 0.003 and -0.001, respectively. Thus an economy with the same gravity structure but no endogenous location margin does not mechanically generate a positive relationship between recovered exporter competitiveness and market

access. The Monte Carlo therefore supports the internal consistency of the two-step identification design.

7.2 External Validation: Model-Implied Location Shares versus Data

The internal validation exercise in Section 7.1 shows that, within the model class, the proposed gravity-based procedure recovers γ when the data are generated by the assumed structure. This is a necessary check, but not a sufficient one. I therefore provide an external validation by comparing theory-consistent objects recovered from the estimation procedure with independent data not used in estimation.

The natural object for external validation is $\mu_{i,t}$, the model-implied share of producers that choose country i as their location and hence a key determinant of its industrial base. Among the objects recovered from the model, $\mu_{i,t}$ is most directly tied to the endogenous-location mechanism and admits an independent empirical counterpart. Other recovered objects, such as market access, are latent gravity constructs or aggregates that can arise even in models without endogenous location choice. I therefore validate the mechanism by asking whether the recovered location shares, $\hat{\mu}_{i,t}$, align with independent measures of the producer base.

I construct the external counterpart to $\mu_{i,t}$ from the UNIDO INDSTAT data on the number of manufacturing establishments, which can be considered the closest empirical analogue to the model’s measure of producers. I harmonize ISIC Revision 3 and ISIC Revision 4 into a single panel data sample to maximize coverage. Let $N_{i,t}$ denote the resulting external measure of the number of manufacturing establishments in country i and year t . I convert this object into an empirical location share by normalizing across countries within each year:

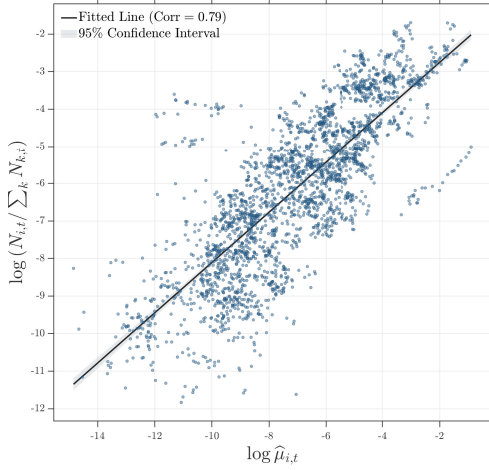
$$\mu_{i,t}^{\text{ext}} = \frac{N_{i,t}}{\sum_k N_{k,t}}.$$

This normalization matches the model object because $\mu_{i,t}$ is the within-period share of the world mass of producers located in country i . The external measure is constructed entirely from data that are neither used nor targeted in the estimation and is therefore suitable for external validation.

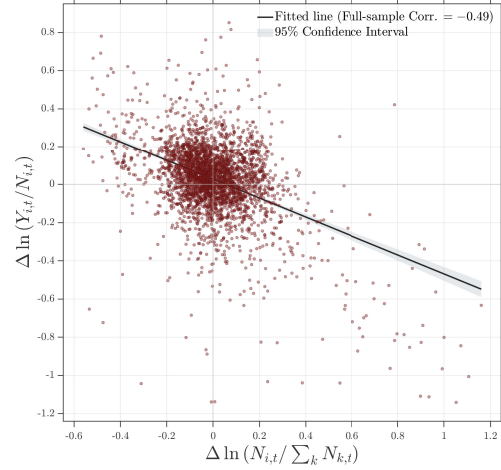
Panel A in Figure 3 plots model-recovered $\mu_{i,t}$ versus externally calculated $\mu_{i,t}^{\text{ext}}$. The relationship is strongly positive, with a correlation of 0.8, suggesting that the mass

of producers predicted by the model and recovered from the gravity estimation captures meaningful variation in the independent establishment-based measure of where producers are located. This relationship provides meaningful external validation of the endogenous-location mechanism, since the recovered production-base variable is constructed from trade flows while the external measure is constructed independently from establishment data.

(a) PANEL A: MASS OF PRODUCERS



(b) PANEL B: REVENUE PRODUCTIVITY



Notes: Panel A compares the recovered model-implied firm-location share, $\hat{\mu}_{i,t}$, with the external establishment-based measure, $N_{i,t} / \sum_k N_{k,t}$. Panel B plots annual changes in average revenue per manufacturing firm, $\Delta \ln(Y_{i,t} / N_{i,t})$, against annual changes in the country's share of the global manufacturing firm base, $\Delta \ln(N_{i,t} / \sum_k N_{k,t})$. Each point is a country-year observation. In both panels, the solid line is the fitted linear relationship, and the shaded area is the 95% confidence interval.

Figure 3: EXTERNAL VALIDATION

Panel B of Figure 3 provides a second external validation exercise. It uses the model prediction that average revenue productivity among firms located in country i declines with the location share $\mu_{i,t}$. From equation (6), the relevant conditional revenue-productivity term is as follows:

$$\mathcal{R}_{i,t} = H_{i,t}^{\eta/\theta} \mu_{i,t}^{-\eta/(\theta\rho)}.$$

Taking log changes gives

$$\Delta \ln \mathcal{R}_{i,t} = \frac{\eta}{\theta} \Delta \ln H_{i,t} - \frac{\eta}{\theta\rho} \Delta \ln \mu_{i,t}.$$

Holding the fundamental component $H_{i,t}$ fixed, this reduces to

$$\Delta \ln \mathcal{R}_{i,t} = -\frac{\eta}{\theta\rho} \Delta \ln \mu_{i,t}.$$

The intuition is that as more firms choose to locate in country i , selection becomes weaker and the marginal entrants are less productive than the firms already located there. Average revenue productivity therefore falls as the domestic firm base expands. To assess this prediction, I use average manufacturing output per firm as an empirical proxy for average revenue productivity and plot its annual log change against the annual log change in the country's share of the global manufacturing firms. To keep the figure legible, observations outside the first and ninety-ninth percentiles of either variable are omitted from the scatterplot, while the reported correlation is computed using the full sample. The correlation is strongly negative at -0.49 , consistent with the model's prediction that expansions of the firm base are accompanied by lower average productivity among located firms.

7.3 Quantifying the Welfare Gains

The estimation procedure recovers β and γ . The former identifies the trade elasticity through $\eta = -\beta - 1$, while the latter is defined as $\gamma = \left(\frac{\theta\rho}{\eta} - 1\right)$. As it turns out, combined with η this term is sufficient to calculate the welfare gains from trade without ever separating ρ because the formula in Proposition 2 can be reformulated as:

$$\Delta \ln W_i = \underbrace{-\frac{1}{\eta} \Delta \ln \pi_{ii}}_{\text{domestic share}} + \underbrace{\frac{\gamma}{\eta(1+\gamma)} \Delta \ln \mu_i}_{\text{industrial base}} - \underbrace{\Delta \ln \Lambda_i}_{\text{tariff wedge}}.$$

Hence, the welfare changes are a function of observables $\{\Delta \ln \pi_{ii}, \Delta \ln \mu_i, \Delta \ln \Lambda_i\}$ and parameters $\{\eta, \gamma\}$, and they are identical for perfect competition, Bertrand competition, and monopolistic competition.

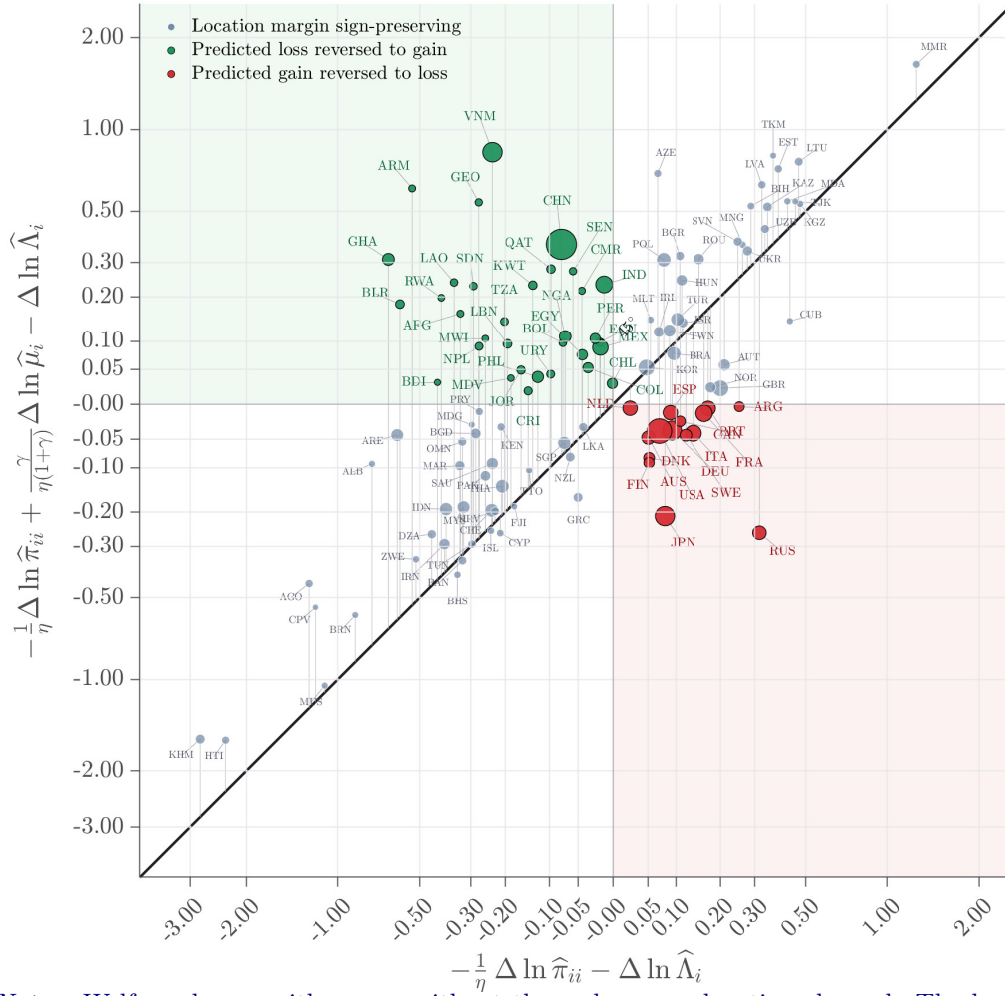
I compute the welfare gains and the three respective components using long difference between 1992 and 2019 for 113 countries for which the data are available in that period. The international trade data and tariff data are used to calculate changes in the domestic expenditure shares and tariff wedges directly; the industrial-base share $\hat{\mu}_i$ relies on the recovered fixed effects and market access as $\hat{\mu}_{i,t} = \exp(\hat{\delta}_{i,t} + \ln \hat{\mathcal{M}}_{i,t}) / \sum_{\ell} \exp(\hat{\delta}_{\ell,t} + \ln \hat{\mathcal{M}}_{\ell,t})$ validated in Section 7.2.

The results are presented in Figure 4. The figure is designed to elucidate the difference that the industrial base term brings to otherwise standard welfare calculations. For that, I plot full welfare changes over 1992-2019 against those that rely only on the ACR term and the tariff wedge such that the difference reflects exactly the contribution of the industrial base. There are two major findings that stand out.

First, the industrial-base term is quantitatively as important as the conventional sufficient statistics. Across the 113 countries observed in both 1992 and 2019, the average country gained approximately 6% in welfare, while the median country gained approximately 4%. The unweighted mean absolute ACR contribution is 0.27 log points, and the corresponding mean absolute industrial-base contribution is 0.25 log points. The tariff-wedge contribution is substantially smaller, with a mean absolute value of 0.05 log points. For 60 countries, the industrial-base term is larger in absolute value than the ACR term. In the United States, the contraction of the industrial base reduces welfare by 0.11 log points, compared with a combined contribution of 0.07 log points from the ACR and tariff-wedge terms. In China, the expansion of the industrial base raises welfare by 0.44 log points, compared with 0.08 log points loss from the other two channels combined.

Second, the industrial-base margin changes the qualitative welfare results for a large set of countries. Among the 113 countries in the sample, the sign of the measured welfare change reverses for 48 countries. For 33 countries, the standard components predict a welfare loss, but the full measure predicts a gain once the expansion of the industrial base is included. These countries appear in the upper-left region of Figure 4 and include several developing and emerging economies, such as China, Mexico, Peru, and Vietnam. For 15 countries, the standard components predict a gain, but the full measure predicts a loss because the industrial base contracts. These countries appear in the lower-right region and include several developed economies, such as the United States, Japan, France, Germany, and Sweden. Thus, whether international trade expands or erodes the domestic industrial base is central to the welfare calculation.

The industrial-base component can flip the sign of measured gains from trade because it often moves in the opposite direction from the ACR component. When a country loses industrial base, its domestic expenditure share tends to fall, so the ACR term rises. The latter therefore records a gain from greater import penetration, while the industrial-base term records the offsetting welfare loss from the contraction of



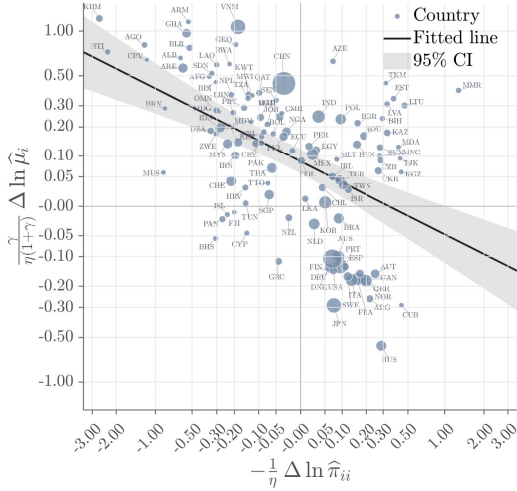
Notes: Welfare change with versus without the endogenous location channel. The horizontal axis omits the industrial-base term, whereas the vertical axis is the full change in welfare. Distance from the 45° line is the industrial-base contribution. Marker size is proportional to 2019 imports and axes use a symlog scale.

Figure 4: WELFARE GAINS FROM TRADE 1992-2019

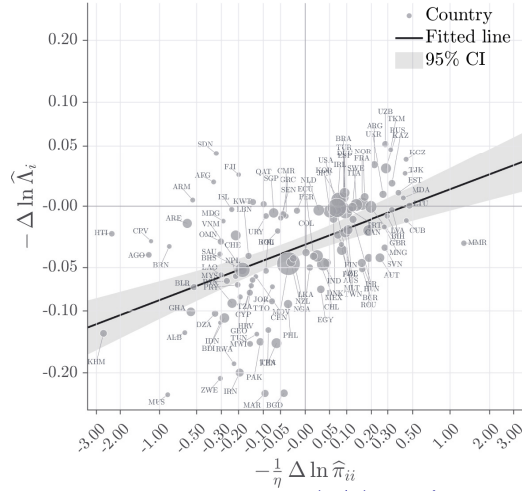
domestic production capacity. This relationship, however, is not mechanical. Some countries may preserve or expand their industrial base while opening to trade and increasing the import share in total expenditure. To examine this pattern, Panel A of Figure 5 plots the industrial-base contribution against the ACR contribution. Panel B plots the ACR contribution against the tariff-wedge contribution.

Panel A confirms a negative correlation between the industrial-base contribution and the ACR contribution. Among the 113 countries observed between 1992 and 2019,

(a) PANEL A: INDUSTRIAL BASE TERM



(b) PANEL B: TARIFF WEDGE TERM



Notes: Horizontal axis in both panels capture trade contribution $-(1/\eta)\Delta \ln \hat{\pi}_{ii}$. Panel A plots the industrial-base contribution $\gamma/(\eta[1+\gamma])\Delta \ln \hat{\mu}_i$. Panel B plots the tariff-wedge contribution $-\Delta \ln \hat{\lambda}_i$. Marker size is proportional to 2019 imports and axes use a symlog scale. The line is the OLS fit in the displayed coordinates with a 95% confidence band.

Figure 5: WELFARE DECOMPOSITION CHANNELS ACROSS COUNTRIES

54 countries increase their domestic expenditure share while also expanding their industrial base. This group includes China and Vietnam. Another 20 countries become more open to trade but lose industrial base, so the ACR contribution is positive while the industrial-base contribution is negative. This group includes several developed economies, such as France, Japan, and the United States. Importantly, greater openness need not come at the expense of the industrial base: 32 countries gain on both margins, with a lower domestic expenditure share and an expanding industrial base including Chile, Colombia, Mexico, Poland, and Turkey.

Panel B plots the tariff-wedge contribution against the trade contribution. The relationship is positive. This is what pure accounting predicts when tariff rates are stable: at positive rates, both terms rise mechanically with import penetration, since domestic expenditure bears no tariff. The positive association therefore indicates that, over this period, movements in the tariff wedge mainly reflect changing import composition at existing rates rather than tariff reductions, which would push the two terms apart. The tariff-wedge contribution thus tends to reinforce the trade term, though it is quantitatively small.

Next, I repeat the decomposition exercise over alternative time windows that correspond to different phases of globalization: the benchmark 1992–2019 period, as well as 2001–2019, 2008–2019, 2012–2019, and 2015–2019. Table 4 decomposes welfare changes into the ACR, industrial-base, and tariff-wedge channels for each horizon. For each period, I report the average welfare change and the cross-country standard deviation. As expected, both statistics decline over shorter horizons.

Table 4: WELFARE DECOMPOSITION IN DIFFERENT TIME PERIODS

Period	N	$\Delta \ln \widehat{W}_i$		Channel contribution			Sign
		Average	Std. Dev.	Trade	Industrial base	Tariff	Reversals
1992–2019	113	+0.058	0.400	−0.087	+0.187	−0.042	41.6%
2001–2019	174	+0.107	0.354	+0.043	+0.077	−0.013	22.4%
2008–2019	175	+0.022	0.301	+0.022	+0.006	−0.007	30.3%
2012–2019	175	−0.029	0.210	−0.028	+0.001	−0.003	26.9%
2015–2019	175	−0.008	0.176	−0.030	+0.024	−0.002	23.4%

Notes: All windows end in 2019. Each row includes all countries observed at both endpoints of the corresponding window, so N varies across rows. Entries are unweighted cross-country means of the welfare change, $\Delta \ln \widehat{W}_i$, and of its three channel contributions, reported in log points. The channels are defined in equation (29) and sum to the total welfare change. “Sign reversals” reports the share of countries for which including the industrial-base term changes the sign of the welfare change implied by the ACR and tariff-wedge terms alone.

The results across different horizons also show that the positive industrial-base contribution is concentrated in the earlier part of the sample. The average industrial-base contribution is 0.187 over 1992–2019 and remains sizable over 2001–2019, at 0.077. After the global financial crisis, however, the average contribution is much smaller: 0.006 over 2008–2019, essentially zero over 2012–2019, and 0.024 over 2015–2019. Thus, most of the positive industrial-base effects in the full-period mean reflect reallocations that accumulated before 2008. This pattern is consistent with the interpretation of the model as comparing long-run equilibria, in which firm-location choices adjust to persistent changes in trade costs and market access.

The results in Table 4 also confirm that the industrial-base channel is qualitatively pivotal in determining the sign of welfare changes across horizons. The industrial-base term reverses the welfare prediction obtained from the ACR and tariff-wedge terms alone for 22.4% of countries over 2001–2019, 30.3% over 2008–2019, 26.9% over 2012–2019, and 23.4% over 2015–2019. The fact that a substantial share of countries experiences sign reversals in every time period shows that the benchmark result is

not specific to a specific period but rather a general result.

7.4 Counterfactual: The Benefits of Foreign Technology

In this section, I return to the counterfactual in [Eaton and Kortum \(2002\)](#) and [Caliendo et al. \(2018\)](#) who study how changes in technology propagate across space through trade. For that, I use the hat algebra approach as in [Dekle et al. \(2007\)](#), writing the equilibrium in proportional changes $\hat{x} = x'/x$ relative to an observed baseline.

I consider an exogenous shock to the technology primitive in China, H_{CHN} . The counterfactual asks how much other countries would gain or lose if Chinese productivity were hit by a negative shock that returned it to its pre-WTO-accession level. To discipline the magnitude of this shock, I use the estimate in [Brandt et al. \(2017\)](#), who find that the tariff reductions associated with China's WTO accession raised Chinese manufacturing productivity by 10.8%. I take this estimate as the target for the counterfactual change in China's technology primitive and study its reversal: how would welfare in the rest of the world change if China's accession-era productivity gains were undone?

In this counterfactual exercise, I hold all other primitives at their observed values. In hat notation, this means setting changes in trade costs, tariffs, labor endowments as follows:

$$\hat{\tau}_{ij} = 1, \quad \hat{\kappa}_{ij} = 1, \quad \hat{L}_i = 1,$$

while holding deficit ratios d_i fixed.

Specifically, the hat-algebra system solves for the equilibrium changes in wages and location shares, $\{\hat{w}_i, \hat{\mu}_i\}$. The baseline expenditure shares π_{ij} , location shares μ_i , bilateral flows X_{ij} , and levels Y_i and E_j are taken from the benchmark equilibrium. I also use the baseline export shares:

$$s_{ij} \equiv \frac{X_{ij}}{Y_i}, \quad \sum_j s_{ij} = 1. \quad (36)$$

To express the counterfactual system in terms of the estimated parameters η and γ , define the composite technology shock as $\hat{A}_i \equiv \hat{H}_i^{\eta/\theta}$. This is the technology shifter that enters trade shares and price indices. Because the primitive productivity scale

is proportional to $H_i^{1/\theta}$, undoing China's 10.8% productivity gain implies:

$$\widehat{H}_{\text{CHN}} = 1.108^{-\theta}.$$

Therefore,

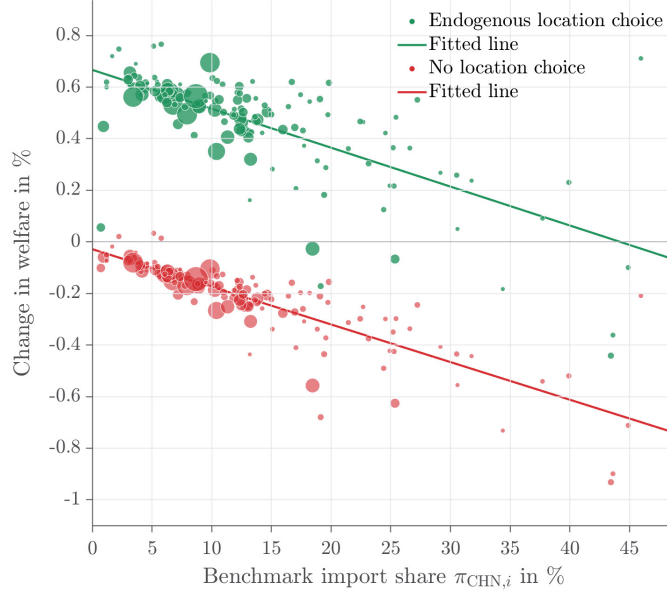
$$\widehat{A}_{\text{CHN}} = \widehat{H}_{\text{CHN}}^{\eta/\theta} = 1.108^{-\eta}, \quad \widehat{A}_i = 1 \quad \text{for } i \neq \text{CHN}.$$

Given these baseline objects and the counterfactual shocks, the counterfactual equilibrium solves conditions (i)–(vii):

$$\begin{aligned} (i) \quad \widehat{\pi}_{ij} &= \frac{\widehat{A}_i \widehat{\mu}_i^{\gamma/(1+\gamma)} (\widehat{w}_i \widehat{\tau}_{ij} \widehat{\kappa}_{ij})^{-\eta}}{\widehat{P}_j^{-\eta}} \\ (ii) \quad \widehat{P}_j^{-\eta} &= \sum_k \pi_{kj} \widehat{A}_k \widehat{\mu}_k^{\gamma/(1+\gamma)} (\widehat{w}_k \widehat{\tau}_{kj} \widehat{\kappa}_{kj})^{-\eta} \\ (iii) \quad \widehat{\mathcal{M}}_i &= \sum_j s_{ij} \widehat{\tau}_{ij}^{-\eta} \widehat{\kappa}_{ij}^{-(\eta+1)} \widehat{P}_j^{\eta} \frac{E'_j}{E_j} \\ (iv) \quad \widehat{\mu}_i &= \frac{\widehat{A}_i^{1+\gamma} \widehat{w}_i^{-\eta(1+\gamma)} \widehat{\mathcal{M}}_i^{1+\gamma}}{\sum_k \mu_k \widehat{A}_k^{1+\gamma} \widehat{w}_k^{-\eta(1+\gamma)} \widehat{\mathcal{M}}_k^{1+\gamma}} \\ (v) \quad \widehat{\Lambda}_i &= \frac{1}{\Lambda_i} \sum_k \frac{\pi_{ki} \widehat{\pi}_{ki}}{\kappa_{ki} \widehat{\kappa}_{ki}} \\ (vi) \quad E'_i &= \frac{(1 + d_i) \widehat{w}_i Y_i}{\widehat{\Lambda}_i \Lambda_i} \\ (vii) \quad \widehat{w}_i &= \frac{1}{Y_i} \sum_j \frac{\pi_{ij} \widehat{\pi}_{ij}}{\kappa_{ij} \widehat{\kappa}_{ij}} E'_j. \end{aligned}$$

To isolate the contribution of the location margin, I compute each country's welfare change twice: once in the full model, in which firms relocate in response to the shock ($\widehat{\mu}_i$ endogenous), and once in a restricted version that holds the location distribution fixed ($\widehat{\mu}_i = 1$), which corresponds to the sufficient-statistic computation standard in the prior literature. The difference between the two is the industrial-base contribution.

Figure 6 plots each country's benchmark import share from China against the welfare effect of reversing China's WTO-accession-era productivity gain. The figure highlights the importance of the endogenous-location channel. In the no-location counterfactual, shown in red, the shock lowers welfare for most countries, with larger losses among



Notes: Welfare change from reversing China's productivity gain. Red markers omit the industrial-base term; green markers report the full change. Solid lines are OLS fits. Marker size is proportional to the benchmark shares μ_i .

Figure 6: WELFARE AND CHINA TECHNOLOGY SHOCK

those that initially import more from China. This is the standard foreign-technology channel: a negative productivity shock in China raises the cost of Chinese goods and reduces welfare by worsening access to foreign varieties.

Once firm location is endogenous, however, the welfare effects are largely reversed. The negative shock to China makes it a less attractive production location and reallocates firms toward the rest of the world. For most countries, the resulting expansion of the industrial base more than offsets the loss from higher import prices, generating positive welfare effects in the full model, shown in green. The contrast between the red and green series underscores the central mechanism of the paper: the welfare consequences of foreign productivity growth depend not only on import exposure, but also on the induced reallocation of production across countries.

8 Conclusions

This paper develops a tractable model of endogenous firm location and embeds it in canonical quantitative trade frameworks. Firms draw location-specific productivities from a multivariate distribution and choose where to produce. A country's industrial

base, defined by the mass and productivity distribution of producers located there, is therefore an equilibrium object rather than a primitive. This location margin introduces a new channel for the gains from trade through changes in the domestic industrial base. I estimate the model structurally to quantify the importance of this channel.

The industrial-base margin is quantitatively first-order. Over 1992 to 2019, its contribution is comparable in magnitude to the ACR term, and the two channels often move in opposite directions. Countries whose domestic expenditure shares fall may gain through the ACR term yet lose through a contraction of the industrial base. Conversely, countries that attract production may gain through the industrial-base channel even when the standard terms imply losses. Accounting for endogenous firm location therefore changes not only the magnitude but also the sign of measured welfare effects for many countries, both in the historical decomposition and in the counterfactual analysis.

The broader implication is that trade affects welfare not only through what countries consume, but also through where production takes place. Domestic expenditure shares alone cannot distinguish gains from improved access to foreign goods from losses caused by an erosion of the domestic production base. This distinction is central to interpreting globalization, evaluating foreign productivity shocks, and designing trade and industrial policy. A natural extension is to introduce forward-looking location decisions in a dynamic setting, which I leave for future research.

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Appendix A: Multivariate Fréchet with Logistic Dependence

In this appendix I construct the multivariate Fréchet distribution with logistic (Gumbel–Hougaard) dependence used in the text and prove Proposition 1. Throughout, z_k is the productivity shock at location k , with shape $\theta > 1$, scale $H_k > 0$, and dependence $\rho \geq 1$; the deterministic return components are the wage w_k and market access $\mathcal{M}_k$, and η is the market-access elasticity.

A1: Construction of Multivariate Fréchet with Logistic Dependence

Begin with the marginal law of each shock z_k , Fréchet with shape $\theta > 0$ and scale H_k :

$$F_k(z) = \exp(-H_k z^{-\theta}). \quad (\text{A1})$$

Define $u_k \equiv F_k(z_k)$, uniform on $[0, 1]$ by the probability integral transform, and impose the logistic copula as discussed in Nelsen (2006):

$$C(u_1, \dots, u_K) = \exp\left(-\left[\sum_{j=1}^K (-\log u_j)^\rho\right]^{1/\rho}\right), \quad \rho \geq 1. \quad (\text{A2})$$

At $\rho = 1$ this is the independence copula $\prod_j u_j$ and as $\rho \rightarrow \infty$ it approaches comonotonicity. By Sklar's theorem $F(\mathbf{z}) = C(F_1(z_1), \dots, F_K(z_K))$. Since $-\log u_k = -\log F_k(z_k) = H_k z_k^{-\theta}$, one can establish the following:

$$\sum_{j=1}^K (-\log u_j)^\rho = \sum_{j=1}^K (H_j z_j^{-\theta})^\rho,$$

leading to the joint law used in Proposition 1:

$$F(z_1, \dots, z_K) = \exp\left(-\left[\sum_{j=1}^K (H_j z_j^{-\theta})^\rho\right]^{1/\rho}\right). \quad (\text{A3})$$

A2: Proof of Proposition 1

The idea behind the proof of Proposition 1 maps the firm location problem into a logistic generalized extreme-value model and then applies standard GEV results for choice probabilities and the joint distribution of the maximizing alternative and the maximized payoff.

The productivity vector $\mathbf{z} = (z_1, \dots, z_K)$ follows (2), with $H_j > 0$, $\theta > 1$, and $\rho \geq 1$. Agents choose k^* to maximize

$$V_k = W_k z_k, \quad W_k \equiv w_k^{-1} \mathcal{M}_k^{1/\eta} > 0.$$

A2.1: Log-payoff transformation

Let me reformulate the payoff problem in logs as follows:

$$U_k = \ln V_k = \ln W_k + \ln z_k.$$

This is a random-utility model with deterministic component $v_k = \ln W_k$ and stochastic component $\tilde{\epsilon}_k = \ln z_k$. Recall that the marginal distribution of z_k is Fréchet(H_k, θ) such that the following holds:

$$F_{\tilde{\epsilon}_k}(y) = \Pr(z_k \leq e^y) = \exp\{-\exp(\ln H_k - \theta y)\}.$$

Hence, $\tilde{\epsilon}_k$ follows a type-I extreme-value distribution with location parameter $\psi_k = \ln H_k/\theta$ and common scale $\varsigma = 1/\theta$.

A2.2: Reduction to the logistic GEV model

To reduce the problem to the logistic GEV, let me define the following variables:

$$u_k = \frac{v_k + \psi_k}{\varsigma} = \theta \ln W_k + \ln H_k, \quad \nu_k = \frac{\tilde{\epsilon}_k - \psi_k}{\varsigma}.$$

Then the log payoff function can be rewritten as:

$$U_k = \varsigma(u_k + \nu_k).$$

Since $\varsigma > 0$, maximizing U_k is equivalent to solving the following problem:

$$k^* = \arg \max_k \{u_k + \nu_k\}.$$

For any threshold vector $\mathbf{x} \in \mathbb{R}^K$:

$$\{\nu_j \leq x_j \text{ for all } j\} = \{z_j \leq \exp(\psi_j + \varsigma x_j) \text{ for all } j\}.$$

Evaluating equation (2) at these thresholds gives

$$F_{\nu}(\mathbf{x}) = \exp \left\{ - \left[\sum_j (H_j \exp [-\theta(\psi_j + \varsigma x_j)])^\rho \right]^{1/\rho} \right\}.$$

Combining the identities that $\theta\psi_j = \ln H_j$ and $\theta\varsigma = 1$ allows to express $H_j \exp [-\theta(\psi_j + \varsigma x_j)]$ as e^{-x_j} such that the following holds:

$$F_{\nu}(\mathbf{x}) = \exp \left\{ - \left[\sum_j e^{-\rho x_j} \right]^{1/\rho} \right\}.$$

This is the symmetric logistic GEV distribution:

$$F_{\nu}(\mathbf{x}) = \exp \left\{ -G(e^{-x_1}, \dots, e^{-x_K}) \right\}, \quad \text{where} \quad G(y_1, \dots, y_K) = \left(\sum_j y_j^\rho \right)^{1/\rho},$$

with dependence parameter $1/\rho$ (McFadden 1978; Train 2009). Then, the implied choice probabilities are:

$$\mu_k \equiv \Pr(k^* = k) = \frac{e^{\rho u_k}}{\sum_j e^{\rho u_j}}.$$

Recall that $e^{u_k} = H_k W_k^\theta$ and the expression for the location shares can be rewritten as:

$$\mu_k = \frac{H_k^\rho w_k^{-\theta\rho} \mathcal{M}_k^{\theta\rho/\eta}}{\sum_j H_j^\rho w_j^{-\theta\rho} \mathcal{M}_j^{\theta\rho/\eta}},$$

which proves part (i) of Proposition 1.

A2.3: Conditional distribution of the selected shock

By the standard GEV result established above, $V^* \equiv \max_j V_j$ is independent of the identity of the maximizing location, k^* . Moreover, conditional on $k^* = k$, the winning payoff satisfies $V_k = V^*$ such that the following holds:

$$F_{V_k|k^*=k}(v) = \Pr(V^* \leq v \mid k^* = k) = \Pr \left(z_j \leq \frac{v}{W_j} \text{ for all } j \right) = \exp \left(-\Xi v^{-\theta} \right),$$

where the aggregate term is defined as:

$$\Xi \equiv \left[\sum_j (H_j W_j^\theta)^\rho \right]^{1/\rho}.$$

For $x > 0$, use the identity $V_k = W_k z_k$:

$$F_{z_k|k^*=k}(x) = \Pr(z_k \leq x \mid k^* = k) = \Pr(V_k \leq W_k x \mid k^* = k) = \exp(-\Xi W_k^{-\theta} x^{-\theta}).$$

Using the expressions for μ_k and Ξ this can be rewritten as:

$$F_{z_k|k^*=k}(x) = \exp\left(-H_k \mu_k^{-1/\rho} x^{-\theta}\right),$$

which proves part (ii) of Proposition 1.

A2.4: Conditional expectation of the selected shock

Once the distribution of z_k conditional on $(k^* = k)$ is known to be Fréchet with shape θ and scale $s_k = (H_k \mu_k^{-1/\rho})^{1/\theta} = H_k^{1/\theta} \mu_k^{-1/(\theta\rho)}$, it immediately follows that for $\theta > 1$:

$$\mathbb{E}[z_k \mid k^* = k] = s_k \Gamma(1 - \frac{1}{\theta}) = H_k^{1/\theta} \mu_k^{-1/(\theta\rho)} \Gamma(1 - \frac{1}{\theta}),$$

which is part (iii) of Proposition 1. □

Appendix B: Proof of Proposition 2

This appendix proves the results in Proposition 2. The real wage can be expressed in terms of the domestic expenditure share. With $\tau_{ii} = \kappa_{ii} = 1$, equations (20) and (22) imply:

$$\pi_{ii} = \mathcal{A} H_i^{\eta/\theta} \mu_i^{1-\eta/(\theta\rho)} \left(\frac{P_i}{w_i} \right)^\eta.$$

Solving for w_i/P_i leads to the following equations:

$$\frac{w_i}{P_i} = \mathcal{A}^{1/\eta} H_i^{1/\theta} \mu_i^{1/\eta-1/(\theta\rho)} \pi_{ii}^{-1/\eta}. \quad (\text{B1})$$

Substituting (B1) into (28) gives the welfare representation in logs:

$$\ln W_i = -\frac{1}{\eta} \ln \pi_{ii} + \left(\frac{1}{\eta} - \frac{1}{\theta\rho} \right) \ln \mu_i - \ln \Lambda_i + \frac{1}{\theta} \ln H_i + \ln(1 + d_i) + C_m, \quad (\text{B2})$$

where $C_m = \frac{1}{\eta} \ln \mathcal{A} - \ln(1 - \mathcal{S})$ is constant within a market structure. Since d_i is held fixed across equilibria, differencing (B2) leaves only π_{ii} , μ_i , and Λ_i , giving the expression in equation (29).

Appendix C: Multivariate Dagum with Clayton Dependence

In this appendix, I construct the multivariate Dagum distribution with Clayton dependence and prove Proposition 3. Throughout, z_k is the productivity shock at location k , with tail parameter $\xi > 0$, scale $H_k > 0$, and dependence $\delta > 0$; the deterministic return components are the wage w_k and market access $\mathcal{M}_k$, with market-access elasticity η .

C1: Construction of Multivariate Dagum with Clayton Dependence

Let me start with the marginal distribution of each shock z_k , specified as Dagum with parameters $\xi > 0$, $1/\delta > 0$, and $H_k > 0$:

$$F_k(z) = \left[1 + \left(\frac{z}{H_k} \right)^{-\xi} \right]^{-1/\delta}. \quad (\text{C1})$$

Define $u_k \equiv F_k(z_k)$, uniform on $[0, 1]$, and impose the Clayton copula (Nelsen 2006):

$$C(u_1, \dots, u_K) = \left(\sum_{j=1}^K u_j^{-\delta} - (K-1) \right)^{-1/\delta}, \quad \delta > 0. \quad (\text{C2})$$

As $\delta \rightarrow 0$ this is the independence copula; as $\delta \rightarrow \infty$ it approaches comonotonicity. By Sklar's theorem $F(\mathbf{z}) = C(F_1(z_1), \dots, F_K(z_K))$. Since $u_k^{-\delta} = 1 + (z_k/H_k)^{-\xi}$,

$$\sum_{j=1}^K u_j^{-\delta} - (K-1) = K + \sum_{j=1}^K \left(\frac{z_j}{H_j} \right)^{-\xi} - (K-1) = 1 + \sum_{j=1}^K \left(\frac{z_j}{H_j} \right)^{-\xi},$$

which leads to the joint distribution in Proposition 3:

$$F(z_1, \dots, z_K) = \left[1 + \sum_{j=1}^K \left(\frac{z_j}{H_j} \right)^{-\xi} \right]^{-1/\delta}. \quad (\text{C3})$$

C2: Proof of Proposition 3

The productivity vector $\mathbf{z} = (z_1, \dots, z_K)$ follows (C3), with $\xi > 1$, $\delta > 0$, and $H_j > 0$. Firms choose k^* to maximize

$$V_k = W_k z_k, \quad W_k \equiv w_k^{-1} \mathcal{M}_k^{1/\eta} > 0.$$

The proof proceeds in three steps. First, I separate the common dependence component from independent Fréchet shocks. Second, I derive the joint distribution of the selected location and its shock, from which the location shares and the conditional shock distribution follow simultaneously. Third, I recover the conditional productivity distribution.

C2.1: Shock Transformation

To prove the results, I first transform the shocks. Let $\zeta_1, \dots, \zeta_K$ be i.i.d. standard Fréchet random variables with shape parameter ξ :

$$F_\zeta(z) = \exp(-z^{-\xi}), \quad z > 0.$$

Further, let $Q \sim \Gamma(1/\delta, 1)$ be independent of ζ , and define:

$$z_k = H_k Q^{1/\xi} \zeta_k.$$

Conditional on $Q = q$, independence of the ζ_k implies

$$\Pr(z_j \leq e_j \text{ for all } j \mid Q = q) = \exp \left[-q \sum_j \left(\frac{e_j}{H_j} \right)^{-\xi} \right].$$

Integrating over Q leads to the distribution in equation (C3):

$$\Pr(z_j \leq e_j \text{ for all } j) = \left[1 + \sum_j \left(\frac{e_j}{H_j} \right)^{-\xi} \right]^{-1/\delta},$$

and letting all but one argument tend to infinity yields the marginal distribution in (C1).

Because $Q^{1/\xi} > 0$ is common across locations, it drops out of the location choice

problem as follows:

$$k^* = \arg \max_k \{W_k H_k Q^{1/\xi} \zeta_k\} = \arg \max_k \{\phi_k \zeta_k\},$$

where

$$\phi_k \equiv W_k H_k = H_k w_k^{-1} \mathcal{M}_k^{1/\eta}.$$

C2.2: Location shares and conditional selection

The joint probability that location k is selected and its standard shock does not exceed z is as follows:

$$\Pr(\zeta_k \leq z, k^* = k) = \int_0^z \Pr(\phi_j \zeta_j \leq \phi_k t \text{ for all } j \neq k) dF_\zeta(t) = \int_0^z \prod_{j \neq k} F_\zeta\left(\frac{\phi_k}{\phi_j} t\right) dF_\zeta(t).$$

Using that $dF_\zeta(t) = \xi t^{-\xi-1} \exp(-t^{-\xi}) dt$, I can rewrite it as follows:

$$\Pr(\zeta_k \leq z, k^* = k) = \int_0^z \xi t^{-\xi-1} \exp\left[-t^{-\xi} \sum_j \left(\frac{\phi_j}{\phi_k}\right)^\xi\right] dt.$$

Let me define as:

$$A_k \equiv \sum_j \left(\frac{\phi_j}{\phi_k}\right)^\xi = \frac{\sum_j \phi_j^\xi}{\phi_k^\xi}.$$

Then, the following holds:

$$\Pr(\zeta_k \leq z, k^* = k) = A_k^{-1} \exp(-A_k z^{-\xi}).$$

because:

$$\frac{d}{dt} \exp(-A_k t^{-\xi}) = A_k \xi t^{-\xi-1} \exp(-A_k t^{-\xi}),$$

Next, let $z \rightarrow \infty$ to obtain the location share as follows:

$$\mu_k \equiv \Pr(k^* = k) = A_k^{-1} = \frac{\phi_k^\xi}{\sum_j \phi_j^\xi}.$$

Or equivalently, substituting for ϕ_k gives:

$$\mu_k = \frac{H_k^\xi w_k^{-\xi} \mathcal{M}_k^{\xi/\eta}}{\sum_j H_j^\xi w_j^{-\xi} \mathcal{M}_j^{\xi/\eta}},$$

which proves part (i) of Proposition 3.

Next, using $A_k = \mu_k^{-1}$, the joint probability can be rewritten as

$$\Pr(\zeta_k \leq z, k^* = k) = A_k^{-1} \exp(-A_k z^{-\xi}) = \mu_k \exp(-\mu_k^{-1} z^{-\xi}).$$

Dividing by $\Pr(k^* = k) = \mu_k$ gives the conditional distribution

$$F_{\zeta_k | k^* = k}(z) = \Pr(\zeta_k \leq z \mid k^* = k) = \exp(-\mu_k^{-1} z^{-\xi}),$$

which remains Fréchet with shape parameter ξ and exponent-measure coefficient μ_k^{-1} . Defining the conditional scale parameter $\tilde{D}_k \equiv H_k \mu_k^{-1/\xi}$, and following the same derivations as before, the conditional distribution can then be written as:

$$F_{z_k | k^* = k}(x) = \left[1 + \left(\frac{x}{\tilde{D}_k} \right)^{-\xi} \right]^{-1/\delta},$$

which establishes part (ii) of Proposition 3.

C2.4: Conditional expectation of the selected shock

For a Dagum random variable with distribution:

$$F(x) = \left[1 + \left(\frac{x}{D_k} \right)^{-\xi} \right]^{-1/\delta},$$

the m -th moment exists for $m < \xi$ and equals:

$$\mathbb{E}[x^m] = D_k^m \frac{\Gamma(1/\delta + m/\xi) \Gamma(1 - m/\xi)}{\Gamma(1/\delta)}.$$

Because $\xi > 1$, the conditional expectation exists and is

$$\mathbb{E}[z_k \mid k^* = k] = H_k \mu_k^{-1/\xi} \frac{\Gamma(1 - 1/\xi) \Gamma(1/\delta + 1/\xi)}{\Gamma(1/\delta)},$$

which proves part (iii) of Proposition 3. □

Appendix D: Monte Carlo

This appendix provides the details of the Monte Carlo exercise in Section 7.1. To parameterize simulations, I use the estimates recovered from the first-step gravity estimation and by the preferred IV estimate of $\hat{\gamma} = 1.230$. Since the empirical exercise does not separately identify θ/η and ρ , for the Monte Carlo exercise, I fix:

$$\frac{\theta}{\eta} = 1$$

as a normalization. Hence, in this case $\theta = \eta$ in the simulated data generating process. The internal dependence parameter is then set to the following value:

$$\rho_{\text{DGP}} = 1 + \hat{\gamma} = 2.230,$$

such that the γ in the location-choice economy satisfies:

$$\frac{\theta \rho_{\text{DGP}}}{\eta} - 1 = \hat{\gamma}.$$

For simplicity, I normalize $H_i \equiv 1$, hold labor endowments fixed over time, and abstract from tariffs and deficits. Hence, $\Lambda_i = 1$, and final expenditure is:

$$E_i = \frac{w_i L_i}{1 - \mathcal{S}}.$$

I set the common profit share to:

$$\mathcal{S} = \frac{1}{1 + \eta}.$$

Given these value, I draw fundamentals and shocks as follows:

- Labor endowments are drawn once and held fixed across periods:

$$\ln L_i \sim N(0, 0.5^2).$$

- Each country is assigned a fixed location $\ell_i \sim \text{Uniform}([0, 10]^2)$. Bilateral distance is $d_{ij} = \|\ell_i - \ell_j\|$, and the persistent component of trade costs is

$$\overline{\ln \tau_{ij}} = 0.25 \ln d_{ij} \quad \text{for } i \neq j, \quad \overline{\ln \tau_{ii}} = 0.$$

- Trade costs are the only time-varying primitive:

$$\tau_{ij,t} = \exp\{\overline{\ln \tau_{ij}} + \varepsilon_{ij,t}\}, \quad \varepsilon_{ij,t} \sim N(0, 0.20^2),$$

with $\tau_{ii,t} = 1$.

- Observed flows contain multiplicative measurement error:

$$X_{ij,t}^{\text{obs}} = X_{ij,t} \exp(\nu_{ij,t}), \quad \nu_{ij,t} \sim N(0, 0.10^2).$$

I simulate $K = 40$ countries and $T = 25$ periods for each Monte Carlo replication, and repeat the exercise for 100 replications. For each replication, I simulate two economies with the same primitives and the same sequence of trade-cost shocks. In the location-choice economy, the source term is:

$$T_{i,t}^{\text{eff}} = H_i^{\eta/\theta} \mu_{i,t}^{1-\eta/(\theta\rho_{\text{DGP}})}.$$

In the no-location-choice economy, the source term is held fixed at

$$T_{i,t}^{\text{eff}} = H_i^{\eta/\theta}.$$

Thus the no-location economy has the same gravity structure and the same trade-cost shocks as the location-choice economy, but shuts down endogenous reallocation of firms across production locations.

In the first step, I estimate a gravity equation with exporter-year and importer-year fixed effects, imposing the known trade-cost elasticity :

$$\mathbb{E}[X_{ij,t}^{\text{obs}} \mid \cdot] = \exp(\delta_{i,t} + \chi_{j,t} - \eta \ln \tau_{ij,t}).$$

Next, as in the benchmark estimation, the recovered importer-year effects are used to construct market access:

$$\widehat{\mathcal{M}}_{i,t} = \sum_j \tau_{ij,t}^{-\eta} \exp(\widehat{\chi}_{j,t}),$$

and a foreign-market-access instrument based on base-period importer effects as in

the:

$$\widehat{Z}_{i,t} = \sum_{j \neq i} \tau_{ij,t}^{-\eta} \exp(\widehat{\chi}_{j,1}).$$

The second step estimates the structural market-access coefficient:

$$\widehat{v}_{i,t} = \alpha_i + \lambda_t + \gamma \left(\ln \widehat{\mathcal{M}}_{i,t} - \eta \ln w_{i,t} \right) + u_{i,t}.$$

The OLS version uses the recovered market-access term directly. The IV version instruments $\ln \widehat{\mathcal{M}}_{i,t} - \eta \ln w_{i,t}$ with $\ln \widehat{Z}_{i,t} - \eta \ln w_{i,t}$.